

STATIC TIMING ANALYSIS USING PRIME TIME LAB

➤ **Lab_01 – Prime Time Flow:**

- **Task 1:** Restore a Prime Time Session
- **Task 2:** Explore some helpful commands
- **Task 3:** Validate an Existing Prime Time Session
- **Task 4:** Execute the Ru Script and Analyze the run
- **Task 5:** Analyze STA Reports.

Static Timing Analysis (STA)

Static Timing Analysis (STA) is a technique used in digital VLSI design to verify the timing performance of a circuit without applying input test vectors. It checks whether all possible paths in the design meet timing requirements such as setup time, hold time, clock skew, and propagation delay. STA is fast, exhaustive, and works across different process, voltage, and temperature (PVT) corners, making it the industry-standard method for timing verification.

Key Advantages:

- No need for simulation test vectors
- Fast and exhaustive path analysis
- Ensures reliable timing closure before fabrication

PrimeTime Tool

Synopsys PrimeTime is the industry-leading STA tool widely used in the semiconductor industry for timing verification and sign-off. It provides accurate analysis of timing paths using detailed cell libraries (.lib) and design constraints (.sdc). PrimeTime helps identify setup and hold violations, report critical paths, and guide engineers to optimize their design for performance and power.

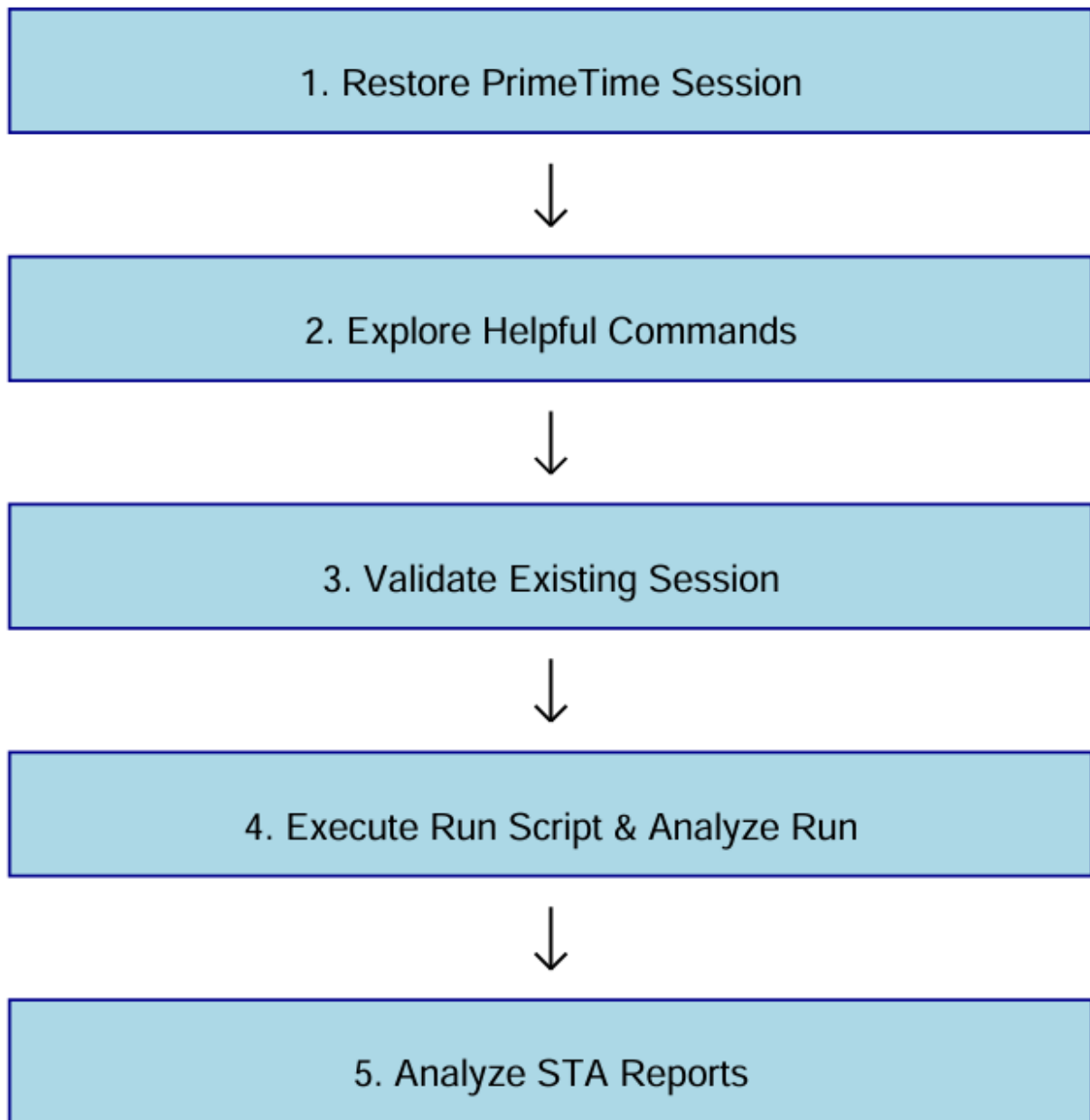
PrimeTime is mainly used for:

- Performing timing sign-off before tape-out
- Checking setup/hold violations
- Generating detailed timing reports
- Supporting multi-corner, multi-mode (MMMC) analysis
- Integrating with physical design tools for ECO (Engineering Change Orders)

Lab 01: PrimeTime Flow:

This lab provides a hands-on introduction to using Synopsys PrimeTime for STA. The focus is on understanding the typical workflow of restoring sessions, running scripts, and analyzing timing results.

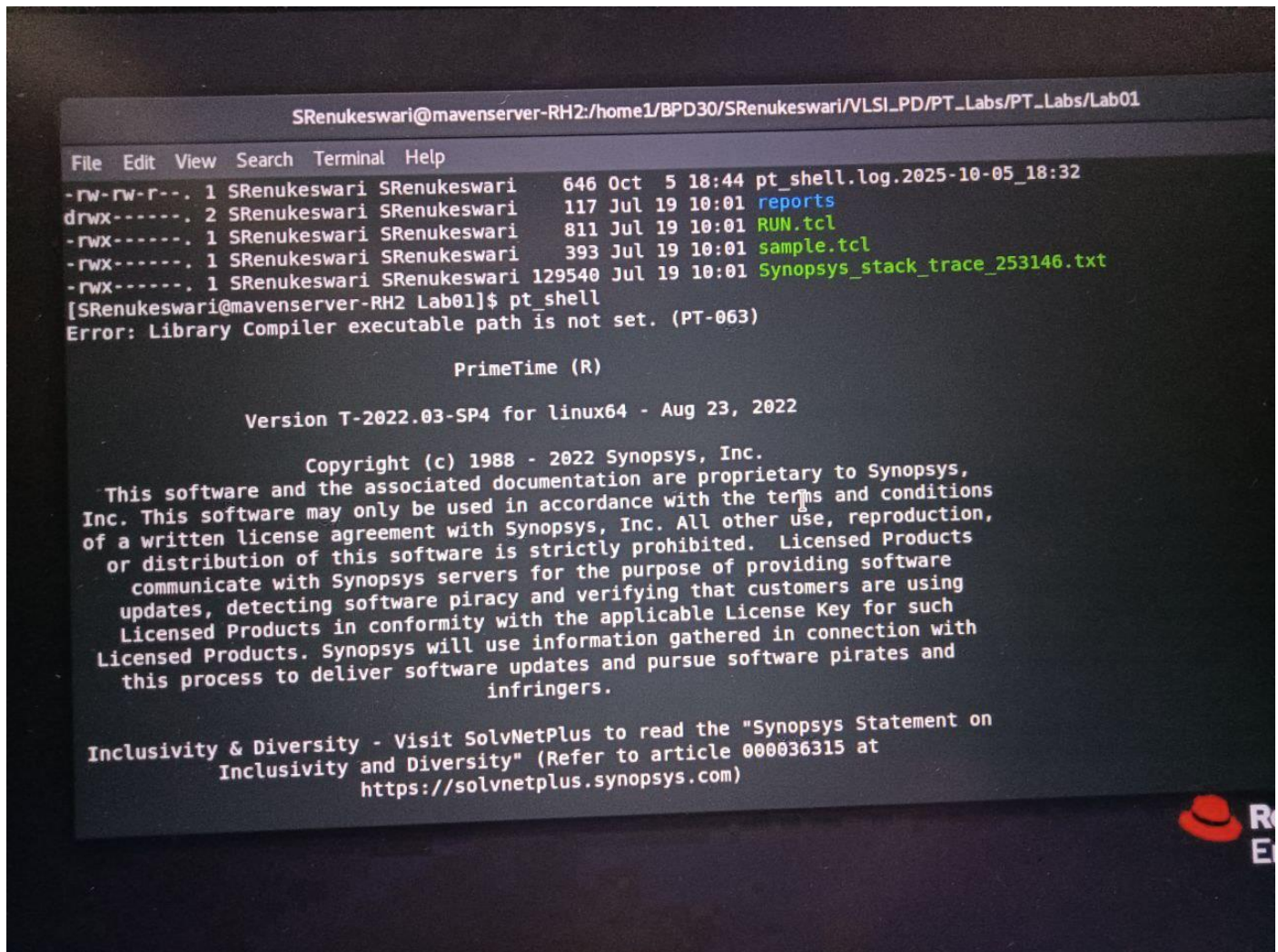
PrimeTime Flow - Static Timing Analysis



Tasks in this Lab:

1. Restore a PrimeTime Session:

- Learn how to reopen a previously saved PrimeTime session.
- Helps continue work without starting from scratch.
- Enter into pt_shell



The screenshot shows a terminal window with the following content:

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
```

File	Edit	View	Search	Terminal	Help	
-rw-rw-r--	1	SRenukeswari	SRenukeswari	646	Oct 5 18:44	pt_shell.log.2025-10-05_18:32
drwx-----	2	SRenukeswari	SRenukeswari	117	Jul 19 10:01	reports
-rwX-----	1	SRenukeswari	SRenukeswari	811	Jul 19 10:01	RUN.tcl
-rwX-----	1	SRenukeswari	SRenukeswari	393	Jul 19 10:01	sample.tcl
-rwX-----	1	SRenukeswari	SRenukeswari	129540	Jul 19 10:01	Synopsys_stack_trace_253146.txt

```
[SRenukeswari@mavenserver-RH2 Lab01]$ pt_shell
Error: Library Compiler executable path is not set. (PT-063)

PrimeTime (R)

Version T-2022.03-SP4 for linux64 - Aug 23, 2022

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```

In the bottom right corner, there is a red hat icon and the letters "RE".

- Restore a previously saved prime time session

```

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help
Restoring miscellaneous application information.....
Completing timing information.....
1
pt_shell> report_analysis_coverage
*****
Report : analysis_coverage
Design : ORCA
Version: T-2022.03-SP4
Date : Mon Oct 6 11:26:46 2025
*****

```

Type of Check	Total	Met	Violated	Untested
setup	9629	3565 (37%)	23 (0%)	6041 (63%)
hold	9629	3535 (37%)	53 (1%)	6041 (63%)
recovery	1316	1210 (92%)	0 (0%)	106 (8%)
removal	1316	1210 (92%)	0 (0%)	106 (8%)
min_period	20	20 (100%)	0 (0%)	0 (0%)
min_pulse_width	7273	5957 (82%)	0 (0%)	1316 (18%)
clock_gating_setup	33	33 (100%)	0 (0%)	0 (0%)
clock_gating_hold	33	33 (100%)	0 (0%)	0 (0%)
out_setup	75	43 (57%)	32 (43%)	0 (0%)
out_hold	75	75 (100%)	0 (0%)	0 (0%)
All Checks	29399	15681 (53%)	108 (0%)	13610 (46%)

```

1
pt_shell> current_design

```

- Check current design by giving current_design command , along with that generate global timing report by giving report_global_timing command.

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/L
File Edit View Search Terminal Help
All Checks 29399 15681 ( 53%) 108 ( 0%) 13610 ( 46%)

1
pt_shell> current_design
{"ORCA"}
pt_shell> report_global_timing
*****
Report : global_timing
        -format { narrow }
Design : ORCA
Version: T-2022.03-SP4
Date   : Mon Oct  6 11:27:34 2025
*****

Setup violations
-----
      Total reg->reg  in->reg  reg->out  in->out
-----
WNS      -3.66      -0.77      -0.22      0.00      -3.66
TNS     -64.59      -3.14      -2.99      0.00     -58.47
NUM        39         9        14         0        16
-----

Hold violations
-----
      Total reg->reg  in->reg  reg->out  in->out
-----
```

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs
File Edit View Search Terminal Help
Report : global_timing
        -format { narrow }
Design : ORCA
Version: T-2022.03-SP4
Date   : Mon Oct  6 11:27:34 2025
*****

Setup violations
-----
      Total reg->reg  in->reg  reg->out  in->out
-----
WNS      -3.66      -0.77      -0.22      0.00      -3.66
TNS     -64.59      -3.14      -2.99      0.00     -58.47
NUM        39         9        14         0        16
-----

Hold violations
-----
      Total reg->reg  in->reg  reg->out  in->out
-----
WNS      -0.04      -0.04      0.00      0.00      0.00
TNS      -0.96      -0.96      0.00      0.00      0.00
NUM        53        53         0         0         0
-----

1
pt_shell> 
```

Here some questions related to this Task:

Q: What is the name of the design under analysis?

A: ORCA [pt_shell>current_design]

Q: How many setup and hold violations does ORCA have?

A: 23 setup and 53 hold violations

Q: How many reg-reg setup and hold violations are there?

A: 9 reg-reg violations and 53 hold violations.

2. Explore Some Helpful Commands:

- Practice basic PrimeTime commands for navigation, reporting, and session handling.
- Builds familiarity with the tool's command-line environment.
- Execute the three history short cut commands:



```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Lab
File Edit View Search Terminal Help
WNS      -0.04      -0.04      0.00      0.00      0.00
TNS      -0.96      -0.96      0.00      0.00      0.00
NUM       53       53        0         0         0
-----
1
pt_shell>
pt_shell>
pt_shell>
pt_shell> history
1 restore_session orca_savesession
2 report_analysis_coverage
3 current_design
4 report_global_timing
5 history
pt_shell> !!
history
1 restore_session orca_savesession
2 report_analysis_coverage
3 current_design
4 report_global_timing
5 history
6 history
pt_shell> !2
report_analysis_coverage
*****
Report : analysis_coverage
Design : ORCA
```

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Lab
File Edit View Search Terminal Help
5 history
6 history
pt_shell> !2
report_analysis_coverage
*****
Report : analysis_coverage
Design : ORCA
Version: T-2022.03-SP4
Date : Mon Oct 6 11:33:24 2025
*****
Type of Check      Total      Met      Violated      Untested
-----
setup              9629      3565 ( 37%)      23 ( 0%)      6041 ( 63%)
hold              9629      3535 ( 37%)      53 ( 1%)      6041 ( 63%)
recovery          1316      1210 ( 92%)       0 ( 0%)      106 ( 8%)
removal           1316      1210 ( 92%)       0 ( 0%)      106 ( 8%)
min_period         20        20 (100%)       0 ( 0%)       0 ( 0%)
min_pulse_width   7273      5957 ( 82%)       0 ( 0%)      1316 ( 18%)
clock_gating_setup 33        33 (100%)       0 ( 0%)       0 ( 0%)
clock_gating_hold 33        33 (100%)       0 ( 0%)       0 ( 0%)
out_setup         75        43 ( 57%)      32 ( 43%)       0 ( 0%)
out_hold          75        75 (100%)       0 ( 0%)       0 ( 0%)
-----
All Checks        29399     15681 ( 53%)     108 ( 0%)     13610 ( 46%)
1
pt_shell> █
```

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help
out_setup          75          43 ( 57%)          32 ( 43%)          0 ( 0%)
out_hold           75          75 (100%)          0 ( 0%)          0 ( 0%)
-----
All Checks         29399        15681 ( 53%)        108 ( 0%)        13610 ( 46%)
1
pt_shell> !3
current design
{"ORCA"}
pt_shell> !1
restore sessio orca_savesession
Information: Clean up the current session for restore:
removing all loaded designs and libraries in memory.... (SR-004)
Removing design 'ORCA'...
Removing library '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/sc_max.db:cb13fs120_tsmc_max'...
Removing library '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/io_max.db:cb13io320_tsmc_max'...
Removing library '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/special.db:cb13special'...
Removing library '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/mem_max.db:cb13_tsmc_memory_max'...
Date session saved: Fri Oct 3 13:19:29 2025
Restoring variable information.....
Restoring netlist information.....
Loading db file '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/sc_max.db'
Loading db file '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/io_max.db'
Loading db file '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/special.db'
Loading db file '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/mem_max.db'
Restoring environmental constraints.....
Restoring timing information.....
Restoring miscellaneous application information.....
```

- Explore the some key bindings in this.



```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/
File Edit View Search Terminal Help
Restoring timing information.....
Restoring miscellaneous application information.....
Completing timing information.....
1
pt_shell> list_key_bindings

Report : key bindings
Version: T-2022.03-SP4
Date   : Mon Oct  6 11:35:59 2025
*****

Information: pt_shell is currently in emacs editing mode. (CLE-06)

Standard key bindings
CTRL+@      -> Mark cursor position
CTRL+A      -> Move to the start of the current line
CTRL+B      -> Move backward a character
CTRL+C      -> Interrupt character
CTRL+D      -> Delete the character underneath the cursor
CTRL+E      -> Move to the end of the line
CTRL+F      -> Move forward a character
CTRL+H      -> Delete previous character
CTRL+I      -> Automatic completion (tab)
CTRL+J      -> Insert newline
CTRL+K      -> Kill the text from point to the end of the line
CTRL+L      -> Clear the screen, reprinting the current line at the
top
```

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab
File Edit View Search Terminal Help
top
CTRL+M      -> Insert newline
CTRL+N      -> History next event
CTRL+O      -> Terminal flush
CTRL+P      -> History previous event
CTRL+Q      -> Terminal start
CTRL+R      -> Reverse incremental search
CTRL+S      -> Terminal stop
CTRL+T      -> Toggle last two characters
CTRL+U      -> Kill the current line
CTRL+V      -> Quoted insert
CTRL+W      -> Kill area between the cursor and mark
CTRL+X      -> Digit that starts argument
CTRL+Y      -> Yank the top of the kill ring into the buffer at
point
CTRL+Z      -> Terminal suspend
BACKSPACE   -> Delete previous character

Multi-character bindings
META+B      -> Move back to the start of the current or previous
word
META+C      -> Capitalize the current word
META+D      -> Kill from point to the end of the current word, or if
between words, to the end of the next word
META+F      -> Move forward to the end of the next word
META+L      -> Lowercase the current word
META+N      -> Search next history
META+P      -> Search previous history
```

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help
META+F -> Move forward to the end of the next word
META+L -> Lowercase the current word
META+N -> Search next history
META+P -> Search previous history
META+U -> Uppercase the current word
META+W -> Copy area between mark and cursor to cut buffer
META+b -> Move back to the start of the current or previous word
META+c -> Capitalize the current word
META+d -> Kill from point to the end of the current word, or if
between words, to the end of the next word
META+f -> Move forward to the end of the next word
META+l -> Lowercase the current word
META+n -> Search next history
META+p -> Search previous history
META+u -> Uppercase the current word
META+w -> Copy area between mark and cursor to cut buffer
CTRL+X CTRL+X -> Exchange the cursor and mark

Arrow key bindings
down -> History next event
up -> History previous event
left -> Move backward a character
right -> Move forward a character
home -> Move to the start of the current line
end -> Move to the end of the line
1
pt_shell> 
```

- Generate a timing report

```

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01 x
File Edit View Search Terminal Help
right      -> Move forward a character
home       -> Move to the start of the current line
end        -> Move to the end of the line

1
pt_shell> report_timing -group [get_path_group *]
*****
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -group **async_default** **clock_gating_default** **default** PCI_CLK SD
RAM_CLK SD_DDR_CLK SD_DDR_CLKn SYS_2x_CLK SYS_CLK
        -sort_by slack
Design : ORCA
Version: T-2022.03-SP4
Date   : Mon Oct 6 11:59:28 2025
*****

Startpoint: I_ORCA_TOP/I_RESET_BLOCK/sys_2x_rst_n_buf_reg
            (rising edge-triggered flip-flop clocked by SYS_2x_CLK)

```



```

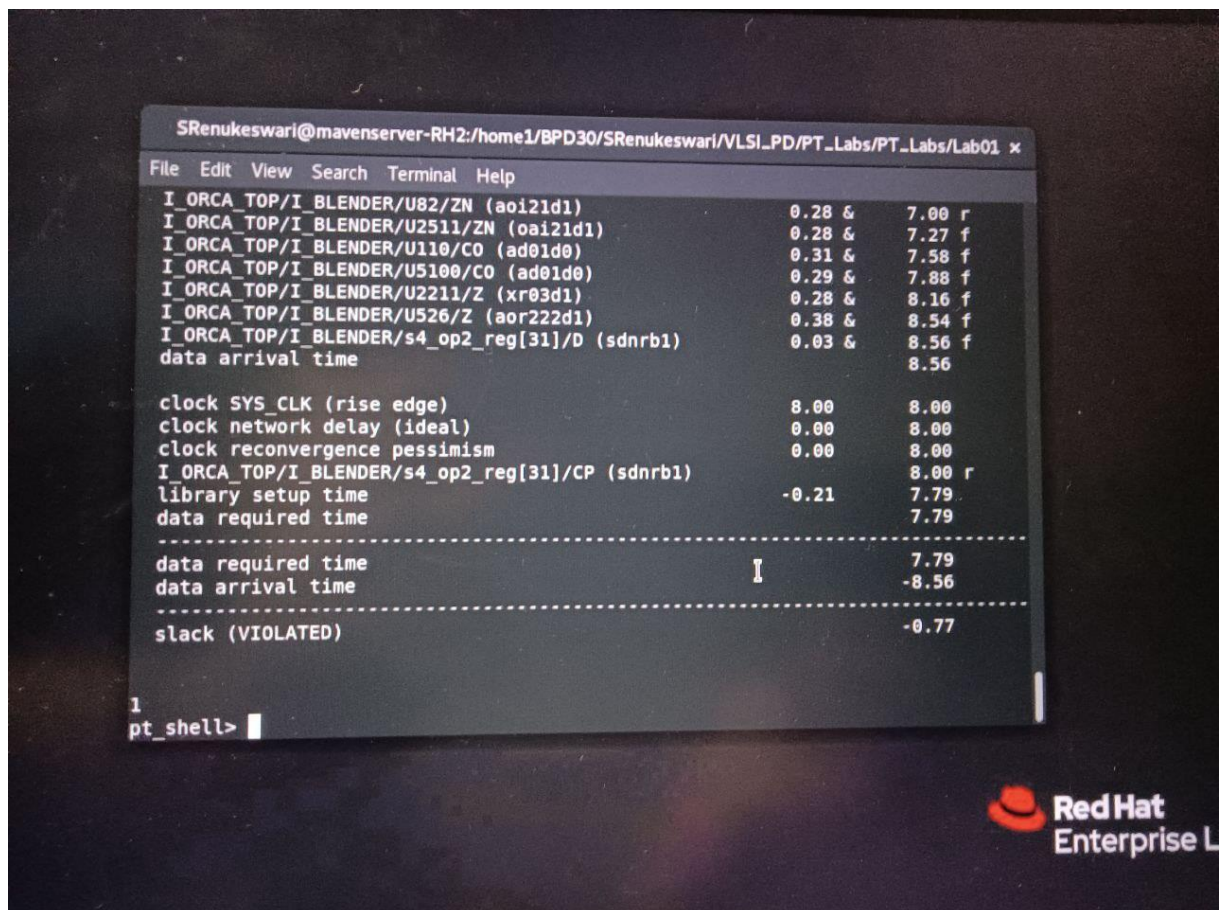
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01 x
File Edit View Search Terminal Help
Date   : Mon Oct 6 11:59:28 2025
*****

Startpoint: I_ORCA_TOP/I_RESET_BLOCK/sys_2x_rst_n_buf_reg
            (rising edge-triggered flip-flop clocked by SYS_2x_CLK)
Endpoint: I_ORCA_TOP/I_RISC_CORE/I_PRGRM_CNT_TOP/I_RISC_CORE_I_PRGRM_CNT_TOP_I
PRGRM_FSM_Current_State_reg[0]
            (recovery check against rising-edge clock SYS_2x_CLK)
Path Group: **async_default**
Path Type: max
Min Clock Paths Derating Factor : 0.90

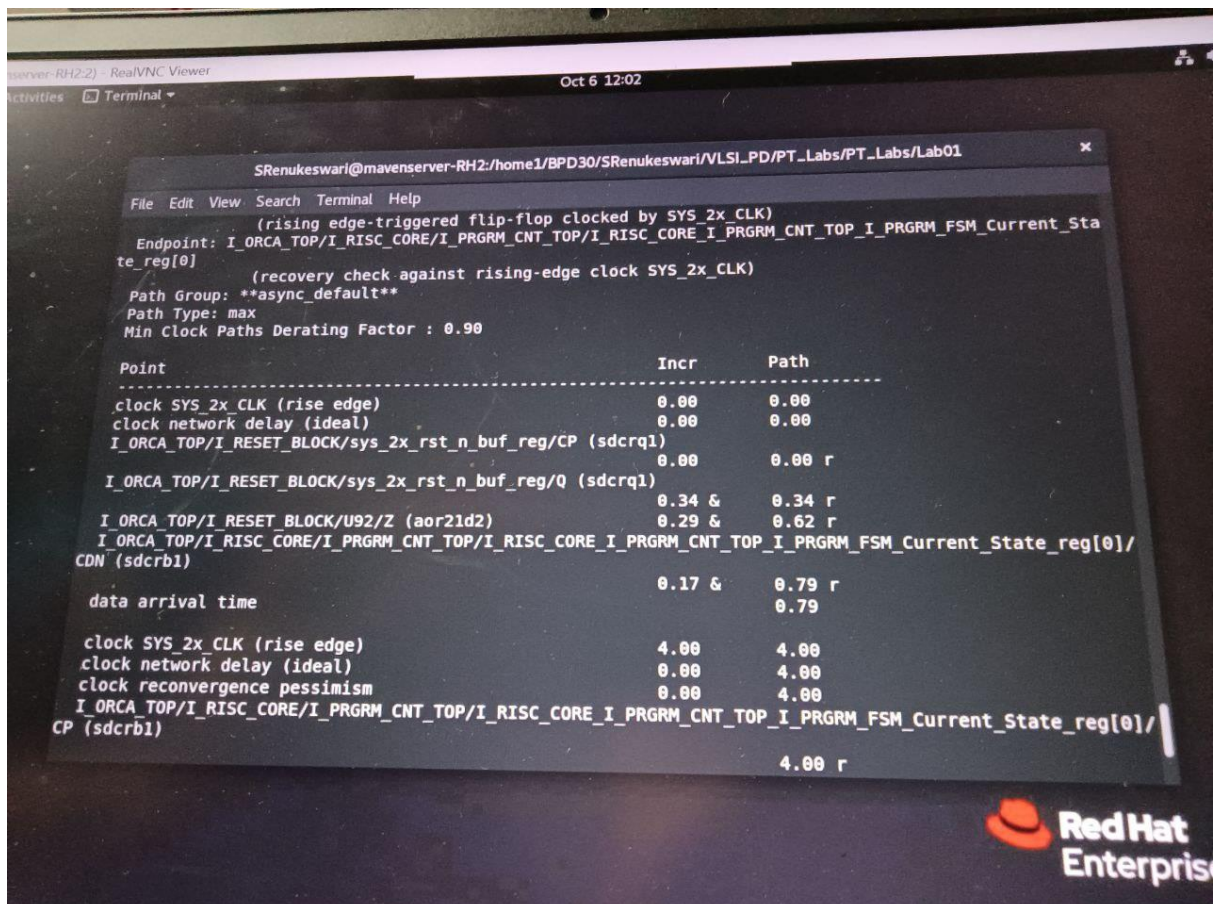
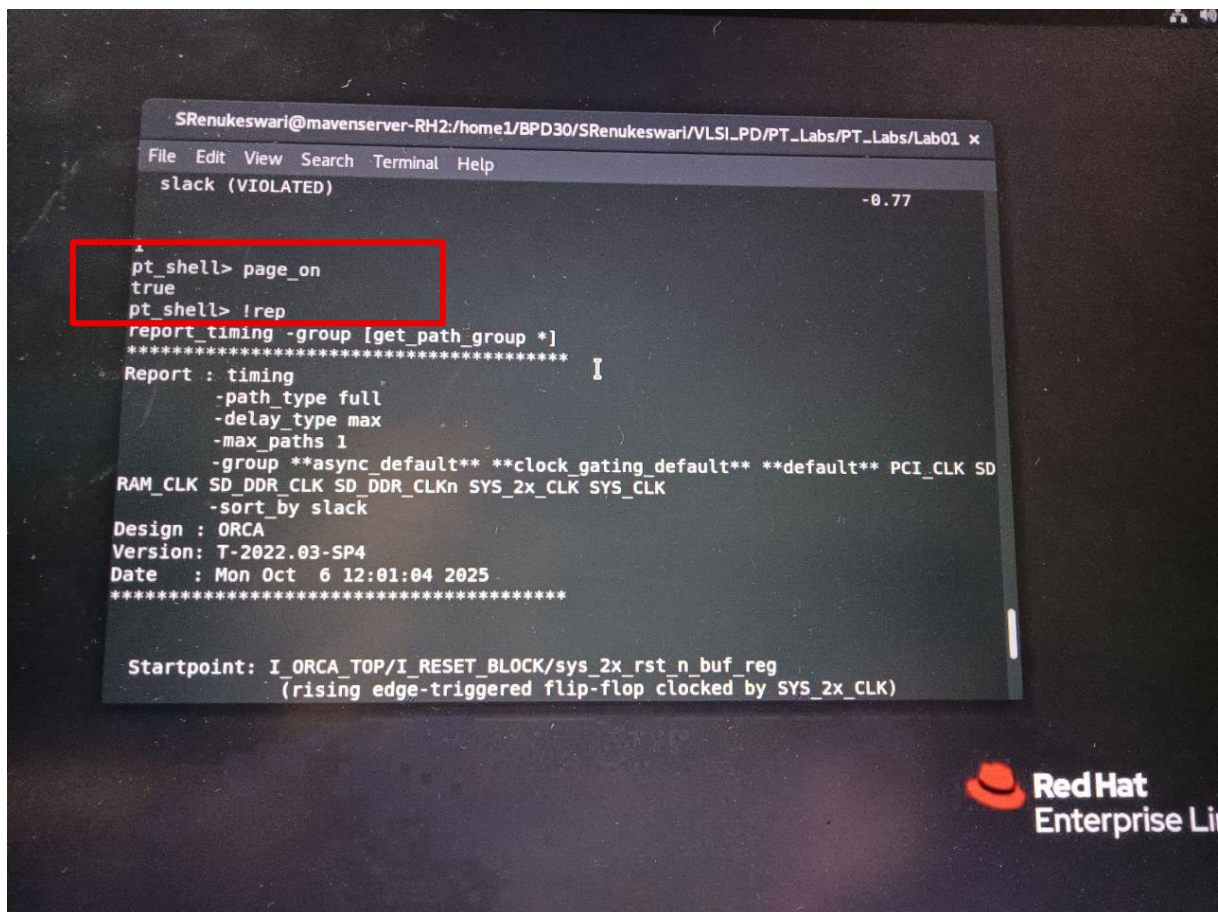
Point                                     Incr      Path
-----
clock SYS_2x_CLK (rise edge)             0.00      0.00
clock network delay (ideal)               0.00      0.00
I_ORCA_TOP/I_RESET_BLOCK/sys_2x_rst_n_buf_reg/CP (sdcrq1) 0.00      0.00 r
I_ORCA_TOP/I_RESET_BLOCK/sys_2x_rst_n_buf_reg/Q (sdcrq1) 0.34 &    0.34 r
I_ORCA_TOP/I_RESET_BLOCK/U92/Z (aor21d2) 0.29 &    0.62 r
I_ORCA_TOP/I_RISC_CORE/I_PRGRM_CNT_TOP/I_RISC_CORE_I_PRGRM_FSM
Current_State_reg[0]/CDN (sdcrb1)

```





- Turn on page mode.

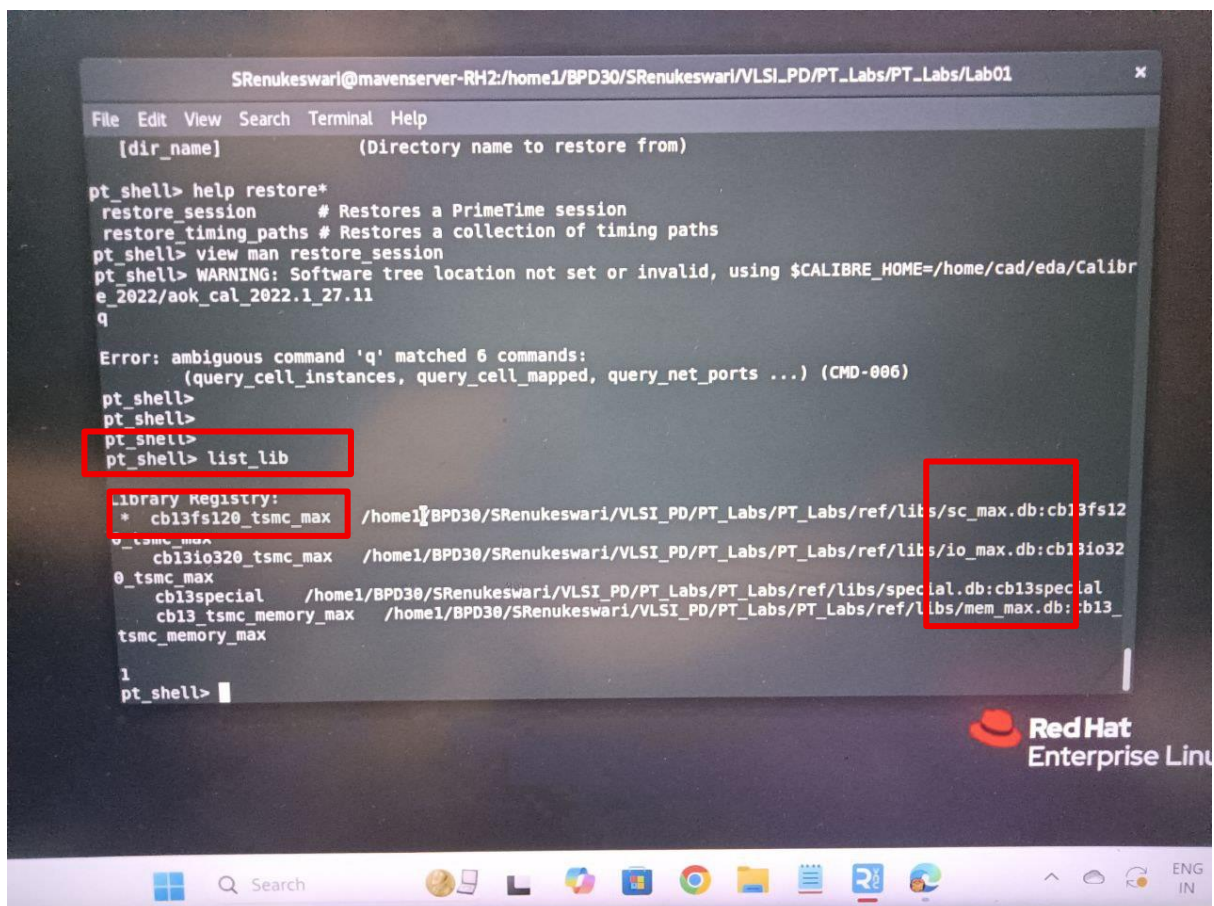
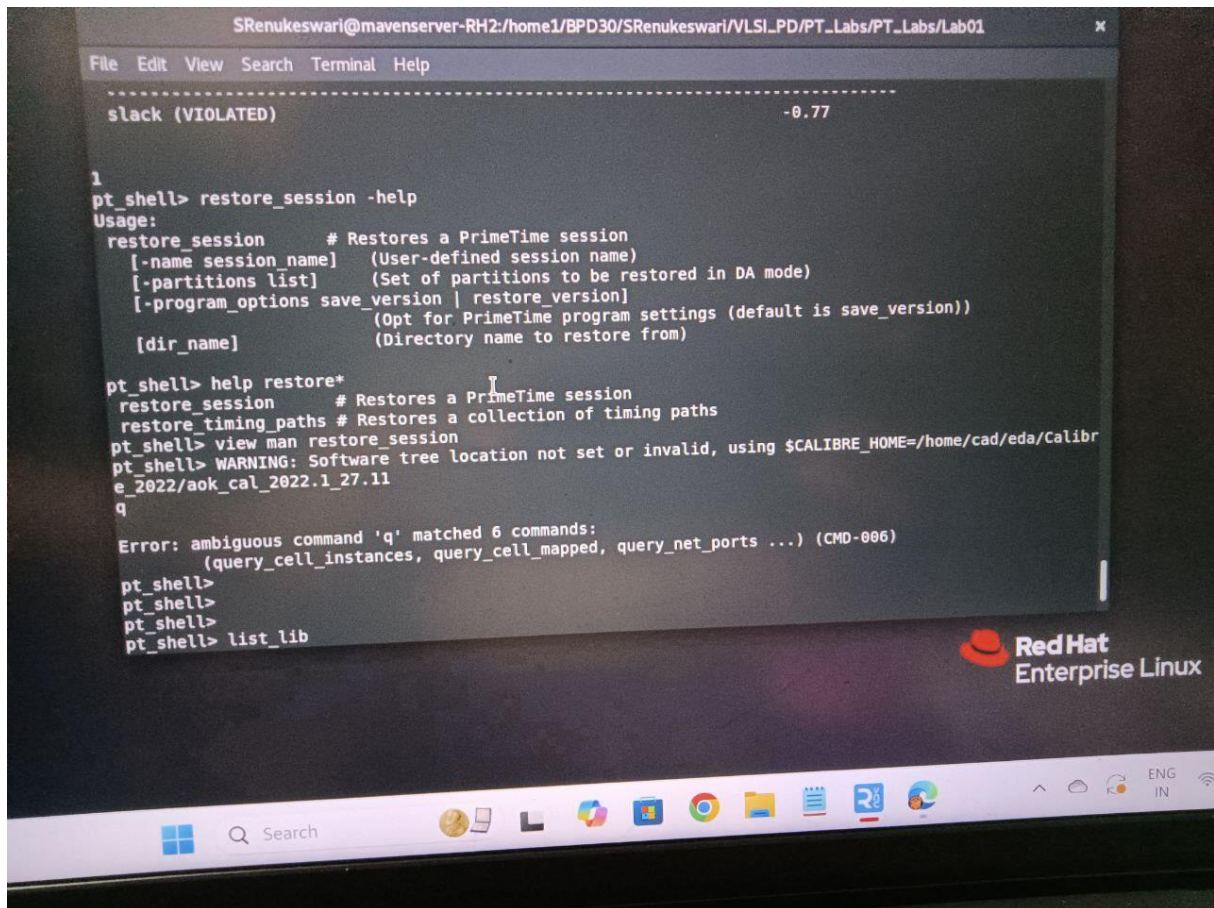


```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help
I_ORCA_TOP/I_BLENDER/U526/Z (aor222d1)      0.38 &      8.54 f
I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]/D (sdnrb1) 0.03 &      8.56 f
data arrival time                             8.56
clock SYS_CLK (rise edge)                    8.00        8.00
clock network delay (ideal)                  0.00        8.00
clock reconvergence pessimism                0.00        8.00
I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]/CP (sdnrb1) 8.00 r
library setup time                          -0.21       7.79
data required time                           7.79
-----
data required time                           7.79
data arrival time                           -8.56
-----
slack (VIOLATED)                             -0.77

1
pt_shell> restore_session -help
Usage:
restore_session    # Restores a PrimeTime session
[-name session_name] (User-defined session name)
[-partitions list]  (Set of partitions to be restored in DA mode)
[-program_options save_version | restore_version]
[dir_name]          (Opt for PrimeTime program settings (default is save_version))
                   (Directory name to restore from)

pt_shell>
```

- Find the command to restore a prime-time session and then display help information on this command.



```

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help
cb13special /home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/special.db:cb13special
cb13_tsmc_memory_max /home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/mem_max.db:cb13
tsmc_memory_max

1
pt_shell> repor_lib cb13fs120_tsmc_max
ERROR: unknown command 'repor_lib' (CMD:006)
pt_shell> report_lib cb13fs120_tsmc_max
*****
Report : library
Library: cb13fs120_tsmc_max
Version: T-2022.03-SP4
Date : Mon Oct 6 12:10:36 2025
*****

Time Unit : 1 ns
Capacitance Unit : 1 pF
Default Wire Load Mode : enclosed
Default Wire Load Model : ForQA
Default Wire Selection Group: predcaps

Operating Conditions:

```

Name	Process	Temp	Voltage	Tree Type
cb13fs120_tsmc_max	1.20	125.00	1.08	worst_case

--More--



```

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help

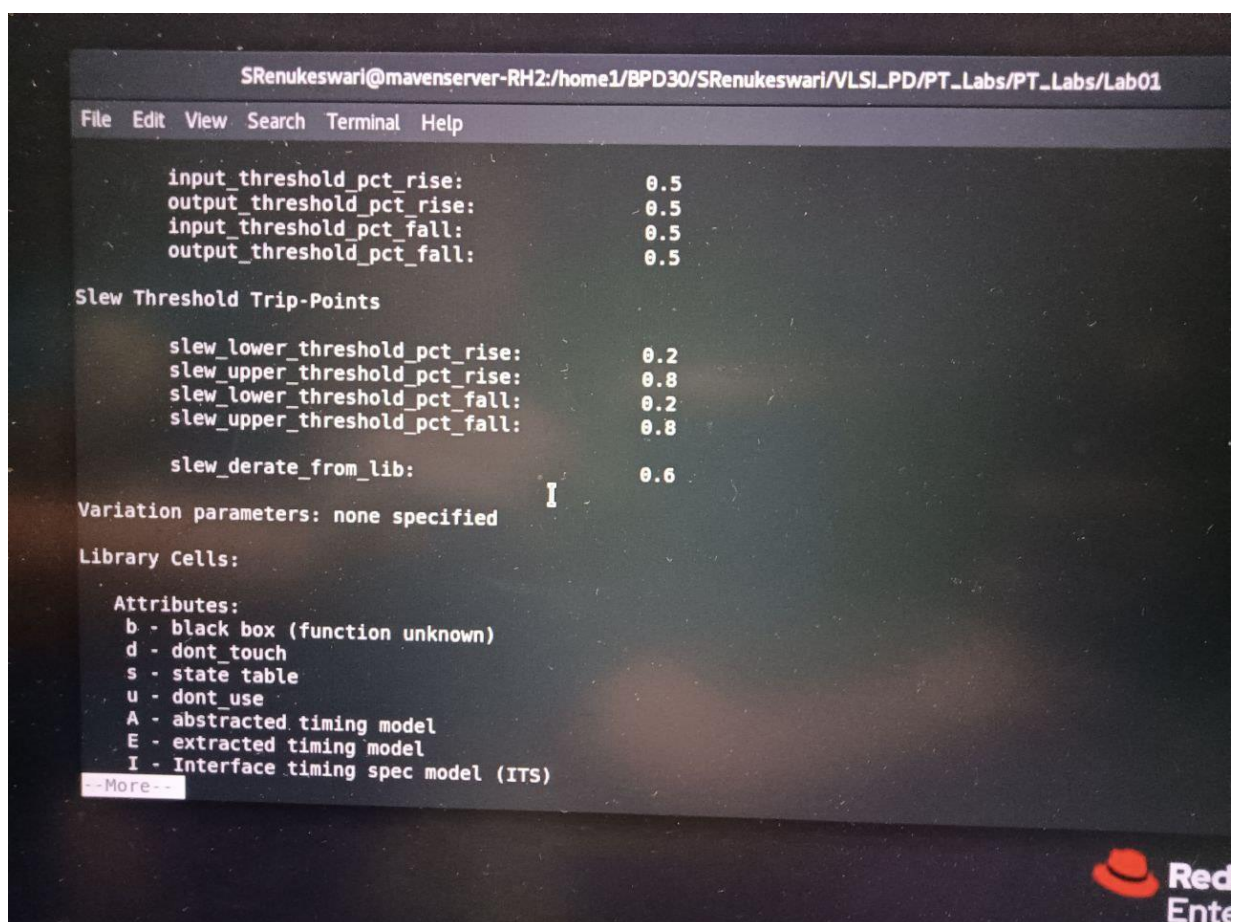
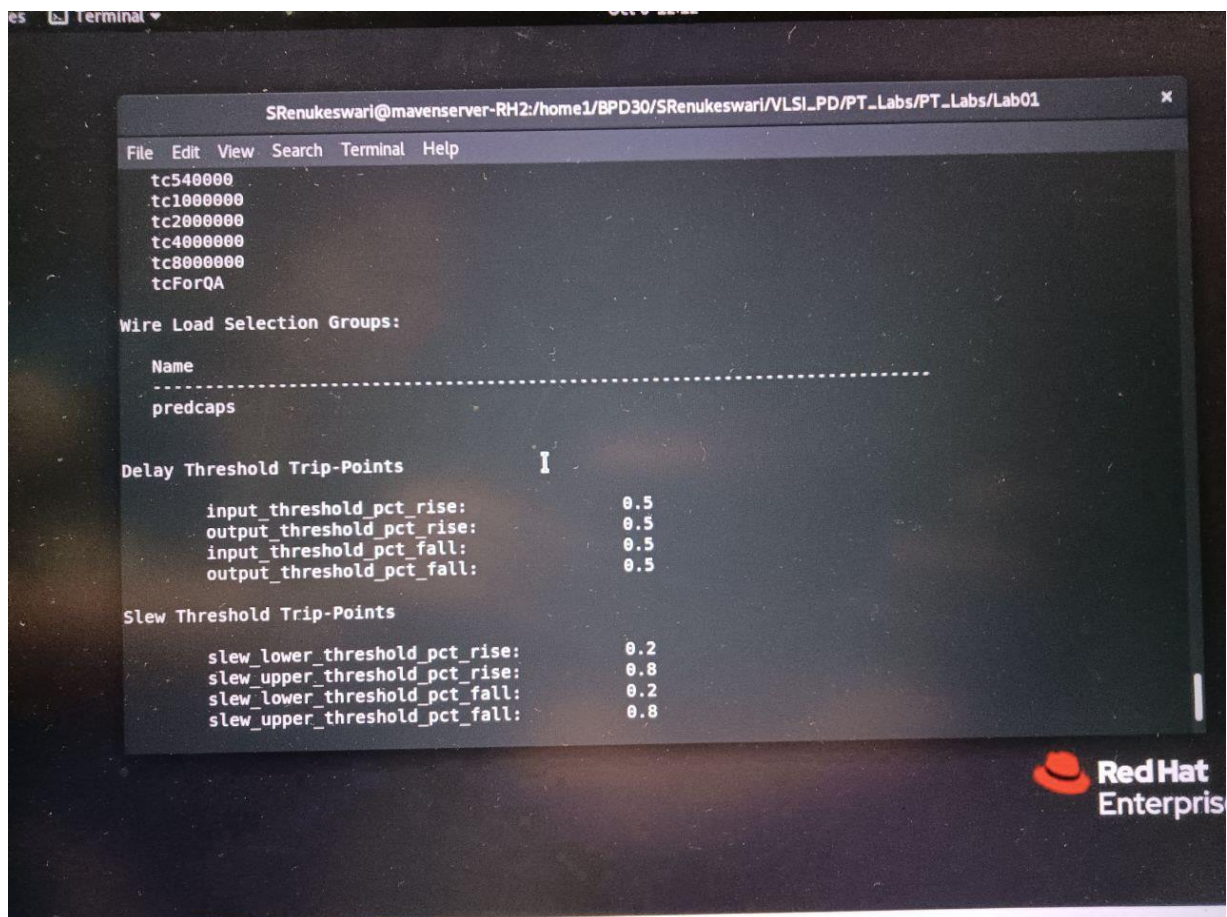
Name Process Temp Voltage Tree Type
-----
cb13fs120_tsmc_max 1.20 125.00 1.08 worst_case

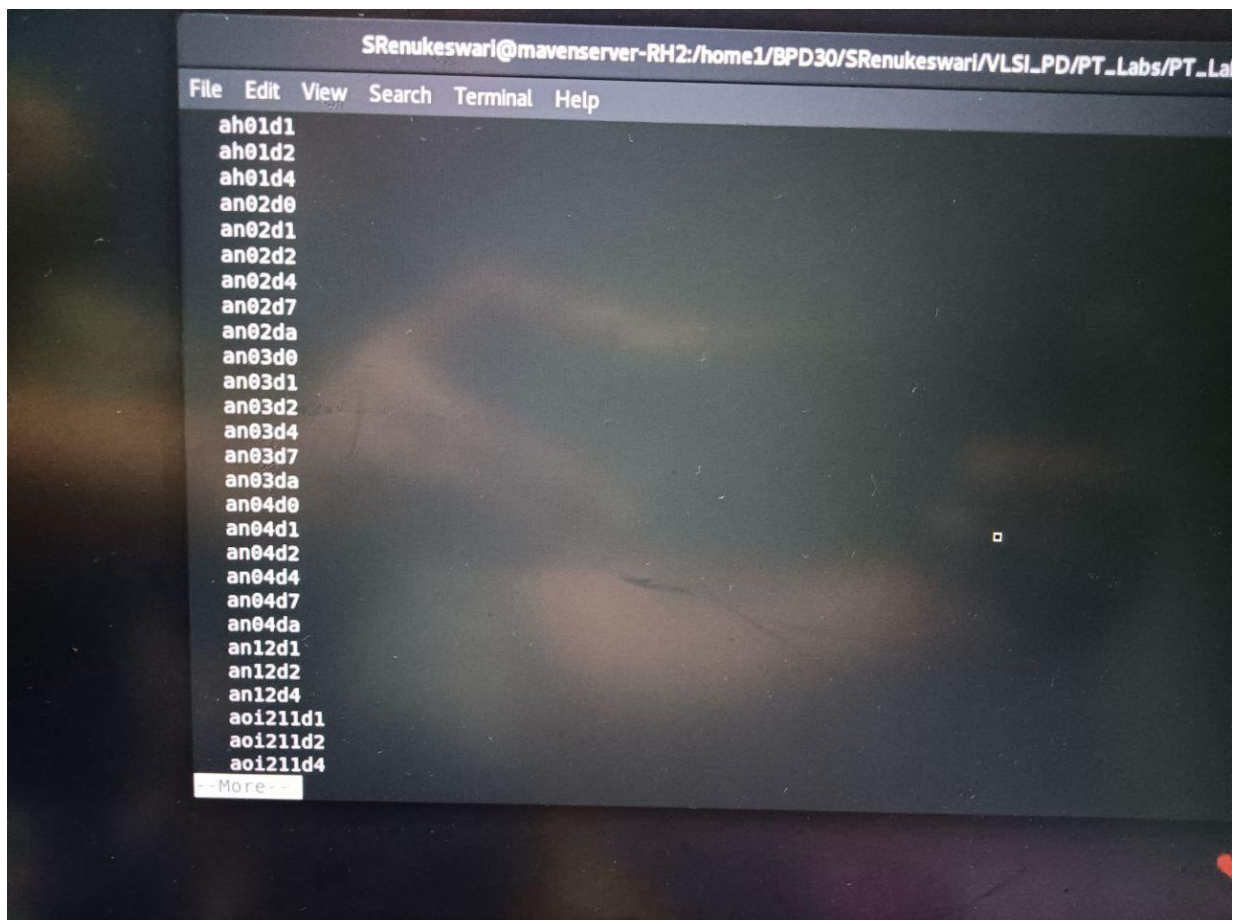
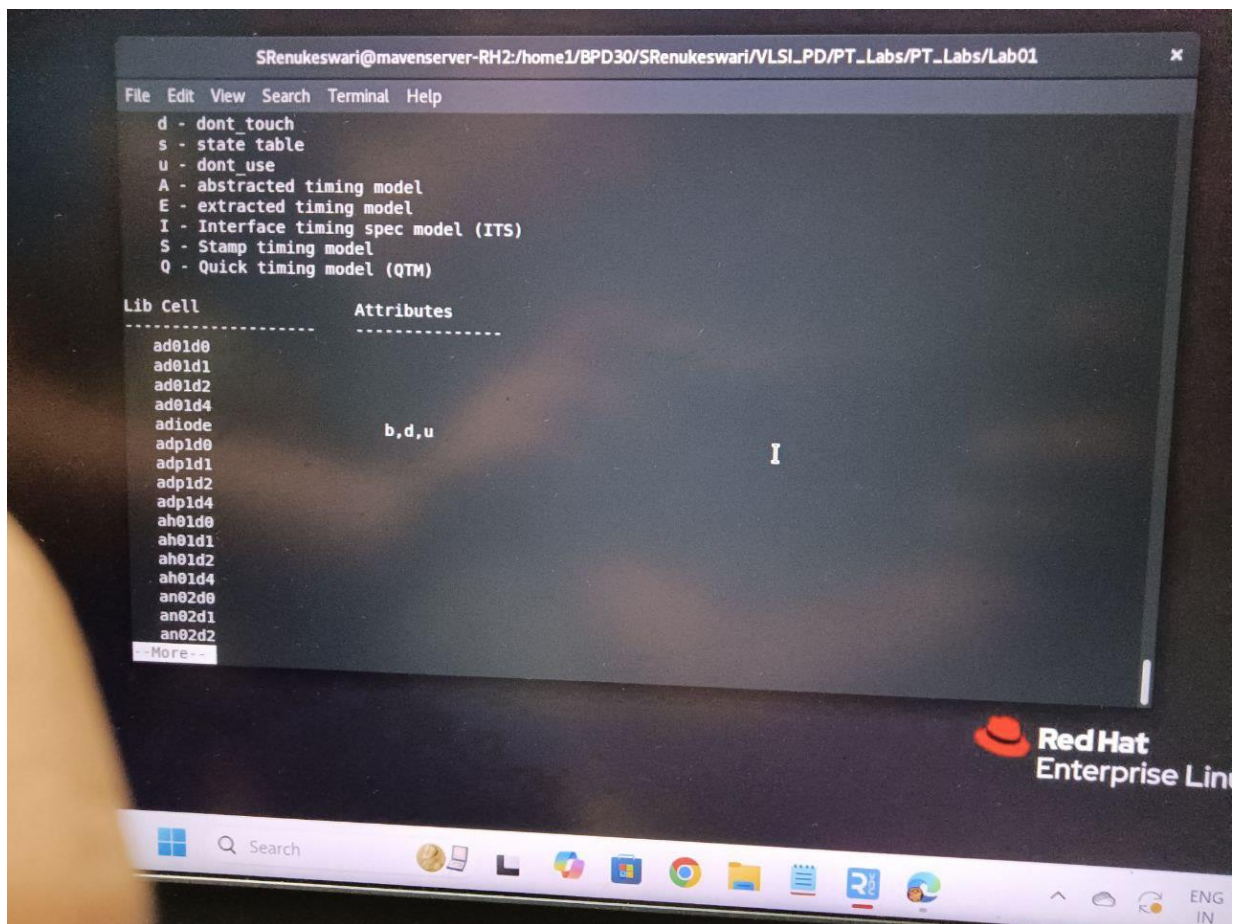
Wire Load Models:
Name
-----
8000
16000
35000
70000
140000
280000
540000
1000000
2000000
4000000
8000000
ForQA
tc8000
tc16000
tc35000
tc70000
tc140000
tc280000

```

--More--







```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help
xr02d2
xr02d4
xr02d7
xr02da
xr03d1
xr03d2
xr03d4
xr03d7
xr03da
1
pt_shell> report_units
*****
Report : units
Design : ORCA
Version: T-2022.03-SP4
Date   : Mon Oct 6 12:16:49 2025
*****
Units
-----
Capacitive_load_unit : 1e-12 Farad
Current_unit         : 1e-06 Amp
Resistance_unit       : 1000 Ohm
Time_unit             : 1e-09 Second
Voltage_unit          : 1 Volt
1
pt_shell>
```

Here some questions Related to this Task:

Q: Describe the difference between the last two history commands above.

A: The command !2 repeats the 2nd command executed in history.

The command !! repeats the last executed command in history.

Q: From the last command above, does the command restore session accent switches?

A: Yes, it accepts a session name allowing a user to specify which of many saved sessions to restore.

i.e., name_session, partition list, program_option save version, dir_name

Q: What is the time unit used for timing reports(as well as all other reports) for the ORCA design?

A: Time unit = 1ns.

3. Validate an Existing PrimeTime Session:

- Ensure that the session loads the correct design, libraries, and constraints.
- Verifies that there are no missing files or mismatched versions.
- Verify the current design is your top-level module:ORCA and compare the unique paths of the libraries.

```
Oct 6 12:19
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01

File Edit View Search Terminal Help
-----
Capacitive_load_unit : 1e-12 Farad
Current_unit          : 1e-06 Amp
Resistance_unit       : 1000 Ohm
Time_unit            : 1e-09 Second
Voltage_unit         : 1 Volt
1
pt_shell> current_design
r*ORCA*1
pt_shell> printvar search_path
search_path = "/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01 /home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/ref/design_data /home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs /home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/ref/scripts /home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/tcl_procs /home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/Lab01/pt_scripts"
pt_shell> printvar link_path
link_path = "sc_max.db io_max.db special.db mem_max.db "
pt_shell> list_libraries
error: unknown command 'list_libraries' (CMD-005)
pt_shell> list_libraries

Library Registry:
* cb13fs120_tsmc_max /home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/sc_max.db:cb13fs120_tsmc_max
  cb13io320_tsmc_max /home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/io_max.db:cb13io320_tsmc_max
  cb13special       /home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/special.db:cb13special
  cb13_tsmc_memory_max /home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/mem_max.db:cb13_tsmc_memory_max
1
pt_shell> 
```

- Verify that the nets are completely annotated.

```
Srenukeswari@mavenserver-RH2:/home1/BPD30/Srenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab0
File Edit View Search Terminal Help
pt_shell> report_annotated_parasitics
*****
Report : annotated_parasitics
-check
-internal_nets
-boundary_nets
Design : ORCA
Version: T-2022.03-SP4
Date : Mon Oct 6 12:21:31 2025
*****
```

Net Type	Total	Lumped	RC pi	RC network	Not Annotated
Internal nets					
- Pin to pin nets	22566	0	0	22566	0
- Driverless nets	24	0	0	0	24
- Loadless nets	112	0	0	112	0
Boundary/port nets					
- Pin to pin nets	71	0	0	0	71
- Driverless nets	0	0	0	0	0
- Loadless nets	0	0	0	0	0
	22773	0	0	22678	95

```
1
pt_shell>
```

```
Srenukeswari@mavenserver-RH2:/home1/BPD30/Srenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab0
File Edit View Search Terminal Help
+-----+
| 22773 | 0 | 0 | 22678 | 95 |
+-----+
1
pt_shell> report_annotated_paeasitics -help
Error: unknown command 'report_annotated_paeasitics' (CMD-005)
pt_shell> report_annotated_parasitics -help
Usage:
report_annotated_parasitics # Report on annotated RC parasitics
[-check] (Check the integrity of RC networks)
[-no_check] (Do not check the integrity of RC networks)
[-internal_nets] (Report on internal nets)
[-boundary_nets] (Report on port nets)
[-driverless_nets] (Show annotation information for only driverless nets)
[-loadless_nets] (Show annotation information for only loadless nets)
[-pin_to_pin_nets] (Show annotation information for only pin-to-pin nets)
[-max_nets num] (Max number of nets reported for the list options)
[-list_annotated] (List net drivers which are backannotated)
[-list_not_annotated] (List net drivers which are not backannotated)
[-ignore_partially_annotated] (Consider partially annotated nets as not annotated)
[-list_lumped] (List lumped nets)
[-constant_arcs] (Keep a separate count for constant nets)
[-via] (Report via)
[-via_verbose] (Report via in details)
[net_list] (Specific nets to report)
pt_shell>
```

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/

File Edit View Search Terminal Help
pt_shell> report annotated_parasitics -list_not annotated
Warning: Nothing implicitly matched 'annotated' (SEL-003)
Error: Nothing matched for object_spec (SEL-005)
0
pt_shell> report annotated_parasitics -list_notannotated
Error: unknown option '-list_notannotated' (SEL-010)
pt_shell> report annotated_parasitics -list_annotated
1. net pc be out[3] (driver: pin 2_ORCA_TOP/I_PCI_CORE/d_out_p_bus_reg[22]/Q)

C in pF      Node      Pin/Port
-----
1e-06        1         --
1e-06        2         --
0.0002       3         --
1e-06        4         --
1e-06        5         --
1e-06        6         --
1e-06        7         d_out_p_bus_reg_22_ASTttcInst760/I (load) [+pin_cap=0.00375pF]
1e-06        8         --
1e-06        9         --
1e-06        10        I_ORCA_TOP/I_PCI_CORE/d_out_p_bus_reg[22]/Q (driver) [+pin_cap=0pF]
1e-06        11        --
1e-06        12        I_ORCA_TOP/I_PCI_CORE/mega_shift_reg[4][12]/SD (load) [+pin_cap=0.00375pF]
-----
0.000211    Total

--More--
```

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_

File Edit View Search Terminal Help
Report : annotated_parasitics
-check
-internal_nets
-boundary_nets
-list_annotated
-max_nets 10
Design : ORCA
Version: T-2022.03-SP4
Date : Mon Oct 6 12:30:01 2025
*****

Net Type      | Total | Lumped | RC pi | RC network | Not Annotated |
-----
Internal nets
- Pin to pin nets | 22566 | 0 | 0 | 22566 | 0 |
- Driverless nets | 24 | 0 | 0 | 0 | 24 |
- Loadless nets | 112 | 0 | 0 | 112 | 0 |
-----
Boundary/port nets
- Pin to pin nets | 71 | 0 | 0 | 0 | 71 |
- Driverless nets | 0 | 0 | 0 | 0 | 0 |
- Loadless nets | 0 | 0 | 0 | 0 | 0 |
-----
| 22773 | 0 | 0 | 22678 | 95 |

1
pt_shell>
```

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab0
File Edit View Search Terminal Help
1
pt_shell> report_annotated_parasitics -list Not_annotated
error: unknown option '-list Not' (CMD-010)
pt_shell> report_annotated_parasitics -driverless_nets
*****
report : annotated_parasitics
        -check
        -internal_nets
        -boundary_nets
        -driverless_nets
Design : ORCA
Version: T-2022.03-SP4
Date   : Mon Oct 6 12:32:33 2025
*****
```

Net Type	Total	Lumped	RC pi	RC network	Not Annotated
Internal nets					
- Driverless nets	24	0	0	0	24
Boundary/port nets					
- Driverless nets	0	0	0	0	0
	24	0	0	0	24

```
1
pt_shell>
```

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Lab
File Edit View Search Terminal Help
| 24 | 0 | 0 | 0 | 24 |
1
pt_shell> report_annotated_parasitics -loadless_nets
*****
report : annotated_parasitics
        -check
        -internal_nets
        -boundary_nets
        -loadless_nets
Design : ORCA
Version: T-2022.03-SP4
Date   : Mon Oct 6 12:33:18 2025
*****
```

Net Type	Total	Lumped	RC pi	RC network	Not Annotated
Internal nets					
- Loadless nets	112	0	0	112	0
Boundary/port nets					
- Loadless nets	0	0	0	0	0
	112	0	0	112	0

```
1
pt_shell>
```

Oct 6 12:35

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01

File Edit View Search Terminal Help

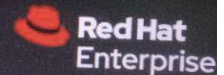
pt_shell> report annotated_parasitics -list_not_annotated

```
1. I_ORCA_TOP/nclk (driver: pin I_ORCA_TOP/nclk)
2. I_ORCA_TOP/test_mem_clk (driver: pin I_ORCA_TOP/test_mem_clk)
3. I_ORCA_TOP/I_RISC_CORE/Test_Regfile_clk (driver: pin I_ORCA_TOP/I_RISC_CORE/Test_Regfile_clk)
4. I_ORCA_TOP/I_RISC_CORE/I_ALU/clk (driver: pin I_ORCA_TOP/I_RISC_CORE/I_ALU/clk)
5. I_ORCA_TOP/I_RISC_CORE/I_INSTRN_LAT/clk (driver: pin I_ORCA_TOP/I_RISC_CORE/I_INSTRN_LAT/clk)
6. I_ORCA_TOP/I_RISC_CORE/I_STACK_TOP/clk (driver: pin I_ORCA_TOP/I_RISC_CORE/I_STACK_TOP/clk)
7. I_ORCA_TOP/I_RISC_CORE/I_DATA_PATH/clk (driver: pin I_ORCA_TOP/I_RISC_CORE/I_DATA_PATH/clk)
8. I_ORCA_TOP/I_RISC_CORE/I_CONTROL/clk (driver: pin I_ORCA_TOP/I_RISC_CORE/I_CONTROL/clk)
9. I_ORCA_TOP/I_RISC_CORE/I_PRGRM_CNT_TOP/clk (driver: pin I_ORCA_TOP/I_RISC_CORE/I_PRGRM_CNT_TOP/clk)
10. I_ORCA_TOP/I_RISC_CORE/I_REG_FILE/clk (driver: pin I_ORCA_TOP/I_RISC_CORE/I_REG_FILE/clk)
..... suppressed 85 additional unannotated nets.
```

```
*****
Report : annotated_parasitics
-check
-internal_nets
-boundary_nets
-list_not_annotated
-max_nets 10
```

```
Design : ORCA
Version: T-2022.03-SP4
Date : Mon Oct 6 12:35:03 2025
*****
```

Net Type	Total	Lumped	RC pi	RC network	Not Annotated
----------	-------	--------	-------	------------	---------------



SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_L

File Edit View Search Terminal Help

Report : annotated_parasitics

```
-check
-internal_nets
-boundary_nets
-list_not_annotated
-max_nets 10
```

```
Design : ORCA
Version: T-2022.03-SP4
Date : Mon Oct 6 12:35:03 2025
*****
```

Net Type	Total	Lumped	RC pi	RC network	Not Annotated
Internal nets					
- Pin to pin nets	22566	0	0	22566	0
- Driverless nets	24	0	0	0	24
- Loadless nets	112	0	0	112	0
Boundary/port nets					
- Pin to pin nets	71	0	0	0	71
- Driverless nets	0	0	0	0	0
- Loadless nets	0	0	0	0	0
	22773	0	0	22678	95

1
pt_shell> █

- Verify the design is completely constrained.

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01 x
File Edit View Search Terminal Help
Boundary/port nets | 0 | 0 | 0 | 0 | 0 |
- Loadless nets | 0 | 0 | 0 | 0 | 0 |
-----+-----+-----+-----+-----+
| 112 | 0 | 0 | 112 | 0 |
|
1
pt_shell> check_timing
Information: Checking 'no_input_delay'.
Information: Checking 'no_driving_cell'.
Information: Checking 'unconstrained_endpoints'.
Warning: There are 2 endpoints which are not constrained for maximum delay.

Information: Checking 'unexpandable_clocks'.
Information: Checking 'latch_fanout'.
Information: Checking 'no_clock'.
Information: Checking 'partial_input_delay'.
Information: Checking 'generic'.
Information: Checking 'loops'.
Information: Checking 'generated_clocks'.
Information: Checking 'pulse_clock_non_pulse_clock_merge'.
Information: Checking 'pll_configuration'.
Information: Checking 'voltage_level'.
0
pt_shell> 
```

- Verify that the checks in your cells are completely exercised; look a possible causes for your findings.

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01 x
File Edit View Search Terminal Help
Information: Checking 'loops'.
Information: Checking 'generated_clocks'.
Information: Checking 'pulse_clock_non_pulse_clock_merge'.
Information: Checking 'pll_configuration'.
Information: Checking 'voltage_level'.
0
pt_shell> report_case_analysis
*****
Report : case_analysis
Design : ORCA
Version: T-2022.03-SP4
Date : Mon Oct 6 12:43:33 2025
*****

Pin name                                User case analysis value
-----
test_mode                                0
scan_en                                  0
power_save                               0
pm66en                                   1
prst_n                                   1

1
pt_shell> 
```

Here some questions related to this Task:

Q: Have 4 libraries in the link_path been successfully read into prime time?

A: Yes, the 4 libraries specified with link_library are the same as those displays with list_libraries.

Q: Which library defines the default for time units, operating conditions, and other delay calculation information?

A: cb13fs120_tsmc_max – the asterisk to the left of the library indicates ‘main library’.

Q: What time unit is used?

A: Time unit = 1ns.

Q: Are there any nets that are not annotated?

A: Yes, there are some internal driverless nets & some boundary pin – to – pin nets that are not annotated.

Q: What options to report_annotated_parasitics would be good to use as a ‘next step’ in debugging the missing nets?

A: report_annotated_parasitics -help shows options; the -list_not_annotated option shows the net names you might want to focus in on. Pin – to – pin nets by using the pin – to – pin_nets options.

Q: What options to check_timing would be good to use as a ‘next step’ in debugging the missing constraints?

A: check_timing -help displays a list of options the -verbose option list the unconstraints endpoints.

Q: Is it logical that many of your timing checks are untested?

A: Yes, test_mode,scan_en,and power_save are constraint off – performing STA on additional modes might enable move checks to be tested.

4. Execute the Run Script and Analyze the Run:

- Run pre-written scripts that automate STA tasks.
- Understand how scripts improve efficiency and consistency in timing analysis.
- Execute the run script logging the result to the log file run.log

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab0
File Edit View Search Terminal Help
SRenukeswari@mavenserver-RH2 Lab01]$ pt_shell -f ./pt_scripts/pt.tcl | tee -i run.log
Error: Library Compiler executable path is not set. (PT-063)
```

PrimeTime (R)

Version T-2022.03-SP4 for linux64 - Aug 23, 2022

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Information: Completed sourcing .synopsys_pt.setup

```
#####
start_profile
1
```

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab0
File Edit View Search Terminal Help
https://solvnetplus.synopsys.com)

Information: Completed sourcing .synopsys_pt.setup

#####
start_profile
1
set timing_update_status_level high
high
#####
set sh_source_emits_line_numbers E
E
set sh_continue_on_error true
true
#####
# PrimeTime Reference Methodology Script
# Script: pt.tcl
# Version: C-2009.06-SP2 (September. 28, 2009)
# Copyright (C) 2008-2009 Synopsys All rights reserved.
#####
### Source common and pt_setup.tcl File ###
###
# Allow source to use search_path
set sh_source_uses_search_path true
true
source ./common_setup.tcl
VDD
```

```

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help
set sh_source_uses_search_path true
true
source ./common_setup.tcl
VDD
VDD
source ./pt_setup.tcl
#####
# Search Path, Library and Operating Condition Section #
#####
set report_default_significant_digits 3 ;
3
set sh_source_uses_search_path true
true
set search_path " $search_path" ;
../ref/design_data ../ref/libs ../ref/scripts ../ref/tcl_procs ../pt_scripts
#####
# Netlist Reading Section #
#####
set link_path " $link_path"
* sc_max.db io_max.db special.db mem_max.db
read_verilog $NETLIST_FILES
1
current design $DESIGN_NAME
Information: current_design won't return any data before link (DES-071)
link design -verbose
Loading verilog file '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/design_data/orca_routed.v.gz'
Loading db file '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/sc_max.db'
Loading db file '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/io_max.db'

```



```

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help
#####
# Back Annotation Section #
#####
read_parasitics $PARASITIC_FILES
Information: Log for 'read_parasitics command' will be generated in 'parasitics_command.log'. (PARA-107)
1
report annotated parasitics -check > $REPORTS DIR/rap.report
#####
# Reading Constraints Section #
#####
if {[info exists CONSTRAINT_FILES]} {
    foreach constraint_file $CONSTRAINT_FILES {
        if {[file extension $constraint_file] eq ".sdc"} {
            read_sdc -echo $constraint_file
        } else {
            source -echo $constraint_file
        }
    }
}
# All application variables for ORCA
# Allow source to use search_path
set sh_source_uses_search_path true
# Report significant digits
set report_default_significant_digits 3
# Terminate script on PrimeTime errors
set sh_continue_on_error false
set sh_script_stop_severity E

```



```

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help
#
# Master constraints for the ORCA chip for functional mode
#
# Time Unit: ns
# Cap Unit : pf
source -echo -verbose orca_pt_clocks.tcl
# Create functional clocks
#####
# PCI clock at 66 MHz
create_clock -period 15 -name PCI_CLK [get_ports pclk]
1
# System clock at 125 MHz
create_clock -period 8 -name SYS_CLK [get_ports sys_clk]
1
# sdram clock at 133 MHz
create_clock -period 7.5 -name SDRAM_CLK [get_ports sdr_clk]
1
#create_clock -p 7.5 -name SD_DDR_CLK [get_pin I_CLOCK_GEN/I_PLL_SD/CLK]
# Create outgoing clocks for DDR interface
#create_generated_clock \
    -source [get_ports sdr_clk] \
    -multiply_by 1 \
    -name SD_DDR_CLK \
    [get_ports sd_CK]
#15 aug 07 -- move source from port to pll pin
create_generated_clock \
    -source [get_pins I_CLOCK_GEN/I_PLL_SD/CLK] \

```

```

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help
    [get_ports sd_CK]
1
#15 aug 07 -- move source from port to pll pin
#create_generated_clock \
    -source [get_ports sdr_clk] \
    -multiply_by 1 -invert \
    -name SD_DDR_CLKn \
    [get_ports sd_CKn]
create_generated_clock \
    -source [get_pins I_CLOCK_GEN/I_PLL_SD/CLK] \
    -multiply_by 1 -invert \
    -name SD_DDR_CLKn \
    [get_ports sd_CKn]
1
# Create internally derived clocks
create_generated_clock \
    -source [get_ports sys_clk] \
    -multiply_by 2 \
    -name SYS_2x_CLK \
    [get_pins I_CLOCK_GEN/I_CLKMUL/CLK_2X]
1
# Setup asynchronous clock groups
set_clock_groups -name func_async -asynchronous \
    -group [get_clocks PCI_CLK] \
    -group [get_clocks SD*] \
    -group [get_clocks SYS*]
1
source -echo -verbose orca_pt_ioconst.tcl

```

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help
1
# Set up constraints for clock gating in Blender design
set_clock_gating_check -setup 0.200 -hold 0.200 [get_clocks SYS_CLK]
1
source -echo -verbose orca_pt_other.tcl
#####
# Other special constraints for ORCA
#####
#####
# Setup case analysis values
set_case_analysis 0 [get_ports "test_mode scan_en power_save"]
1
set_case_analysis 1 [get_ports "pm66en prst_n"]
1
#####
# Small derating factor
set_timing_derate -early 0.9
1
#for RC-005, port is a 1.08 volt, but pad side of pad cell is at 3 volt
create_operating_conditions -name 3V -library cb13i0320_tsmc_max -voltage 3 -process 1 -temperature 1
1
set_operating_conditions 3V -object_list [get_ports *]
1
#for check timing complaint about undriven nets having parasitics
set_driving_cell -lib_cell inv0d1 [all_inputs]
1
#to lower pad cell delays
set_input_transition 0.1 sd_DQ
```

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help
Information: Updating design - Calculating slacks (UITE-214)
Information: Updating design - Calculating slacks (max type) for groups (UITE-214)
Information: Updating design - Calculating slacks (min type) for groups (UITE-214)
Information: Updating design - Completed (UITE-214)
1
#####
# Clock Tree Synthesis Section
#####
set_propagated_clock [all_clocks]
Information: Invalidating logical update. (PTE-139)
1
#####
# Update timing and check timing Section
#####
update_timing -full
Information: Invalidating logical update. (PTE-139)
Information: Abandoning fast timing updates. Forced full update. (PTE-018)
Information: Updating design - Started (UITE-214)
Information: Updating design - Propagating Constants (UITE-214)
Warning: Some timing arcs have been disabled for breaking timing loops or because of constant propagation. Use the 'report_disable_timing' command to get the list of these disabled timing arcs. (PTE-003)
Information: Building multi voltage information for entire design. (MV-022)
Information: Updating design - Calculating delays (UITE-214)
Information: Updating design - Calculating delays 100%... (UITE-214)
Information: Updating design - Calculating delays (UITE-214)
Information: Updating design - Calculating arrivals (UITE-214)
Information: Updating design - Calculating arrivals 10%... (UITE-214)
```

```
Oct 6 12:58

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01

File Edit View Search Terminal Help
Information: Updating design - Calculating slacks (max type) for groups (UITE-214)
Information: Updating design - Calculating slacks (min type) for groups (UITE-214)
Information: Updating design - Completed (UITE-214)
1
# Ensure design is properly constrained
check_timing -verbose > $REPORTS_DIR/ct.report
#####
# Report timing Section
#####
# Report Timing
report_timing -slack lesser than 0.0 -delay min_max -nosplit -input -net -sign 4 > $REPORTS_DIR/rt.report
report_clock -skew -attribute > $REPORTS_DIR/rc.report
report_analysis_coverage > $REPORTS_DIR/rac.report
#####
# Save Session
#####
#####
stop_profile
1
save_session my_savesession
Information: Cleaning and overwriting all data in the existing directory '/home1/BPD30/SRenukeswari/VLSI_PD/PT_La
bs/PT_Labs/Lab01/my_savesession'. (SR-002)
Saving environmental constraints.....
Saving netlist information.....
Saving miscellaneous application information.....
Saving timing information.....
Saving variable information.....
Information: At least 3 MB of free disk space in pt_tmp_dir will be required to restore this session. (SR-044)
```



```
File Edit View Search Terminal Help
PTE-018 Information 10000 - 1 0
SR-002 Information 10000 - 1 0
SR-044 Information 10000 - 1 0
HS-023 Information 100 0 0

Diagnostics summary: 1 error, 2 warnings, 111 informationals
exit
Suppressed Messages Summary:
Id Severity Occurrences Suppressed
-----
RC-009 Warning 478 478
RC-004 Warning 56 56
PTE-071 Warning 32 32
CMD-029 Warning 1 1
CMD-041 Information 54 54
PTE-070 Information 32 32
ENV-003 Information 4 4
Total 7 types of messages are suppressed

Timing updates: 2 (0 implicit, 2 explicit) (0 incremental, 2 full, 0 logical)
Noise updates: 0 (0 implicit, 0 explicit) (0 incremental, 0 full)
Maximum memory usage for this session: 3007.28 MB
CPU usage for this session: 40 seconds
Elapsed time for this session: 15 seconds
Diagnostics summary: 1 error, 2 warnings, 111 informationals

Thank you for using pt_shell!
[SRenukeswari@mavenserver-RH2 Lab01]$
```



SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01

File Edit View Search Terminal Help

Information: At least 3 MB of free disk space in pt_tmp_dir will be required to restore this session

print_message_info

Id	Severity	Limit	Occurrences	Suppressed
PARA-124	Error	100 *	30	0
PT-063	Error	10000 -	1	0
RC-009	Warning	100 *	478	478
RC-004	Warning	100 *	56	56
PTE-071	Warning	10000 -	32	32
PTE-003	Warning	10000 -	2	0
CMD-029	Warning	10000 -	1	1
PARA-007	Warning	100	0	0
UITE-214	Information	10000 -	40	0
PTE-070	Information	100 *	32	32
LNK-044	Information	100 *	24	0
PTE-016	Information	10000 -	12	0
PTE-064	Information	10000 -	12	0
PTE-065	Information	10000 -	8	0
ENV-003	Information	100 *	4	4
LNK-045	Information	10000 -	2	0
MV-022	Information	10000 -	2	0
PARA-107	Information	10000 -	2	0
PTE-139	Information	10000 -	2	0
DES-028	Information	100 *	1	0
DES-071	Information	10000 -	1	0
LNK-034	Information	10000 -	1	0

999 (mavenserver-RH2:1) - RealVNC Viewer

Activities GVim

Oct 6 14:10

tcl_profile_self_only_sorted_by_cpu_time_241...VLSI_PD/PT_Labs/PT_Labs/Lab01/profile) - GVIM

File Edit Tools Syntax Buffers Window Help

Report : Tcl script profile
sorted by cpu time
script levels no higher than at start_profile
Process ID: 2415474
Version: T-2022.03-SP4
Date : Mon Oct 6 12:48:05 2025

Procedure/Command	Calls	Elapsed	CPU	Delta Memory
foreach	1	5	18	0
update_timing	1	4	16	245
link_design	1	0	0	0
report_analysis_coverage	1	0	0	0

"tcl_profile_self_only_sorted_by_cpu_time_2415474.txt" 18L, 887C 1,1 All

bash: cd: tcl_profile_self_only_sorted_by_cpu_time_2415474.txt: Not a directory

[SRenukeswari@mavenserver-RH2 profile]\$ gvim tcl_profile_self_only_sorted_by_cpu_time_241

[SRenukeswari@mavenserver-RH2 profile]\$

```
SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help
1
#####
# Clock Tree Synthesis Section #
#####
set_propagated_clock [all_clocks]
Information: Invalidating logical update. (PTE-139)
1
#####
# Update timing and check timing Section #
#####
update_timing -full
Information: Invalidating logical update. (PTE-139)
Information: Abandoning fast timing updates. Forcing full update. (PTE-018)
Information: Updating design - Started (UITE-214)
Information: Updating design - Propagating Constants (UITE-214)
warning: Some timing arcs have been disabled for breaking timing loops or because of const
e the 'report_disable_timing' command to get the list of these disabled timing arcs. (PTE-
Information: Building multi voltage information for entire design. (MV-022)
Information: Updating design - Calculating delays (UITE-214)
Information: Updating design - Calculating delays 100%... (UITE-214)
Information: Updating design - Calculating delays (UITE-214)
Information: Updating design - Calculating arrivals (UITE-214)
Information: Updating design - Calculating arrivals 10%... (UITE-214)
Information: Updating design - Calculating arrivals 20%... (UITE-214)
```

Here some questions related to this Task:

Q: Were there any error during the execution of the run scripts?

A: No. [Note: Any PARA-124 error during the execution of RUN.tcl(within the parasitics_command.log file) can be safely ignored for the purpose of our labs.

Q: What step required the most CPU time?

A: Update timing – 16 CPU time.

Q: What commands were causing UITE-214 message?

A: Update timing – full

Q: Can any of the timing updates be avoided?

A: Yes, there is an update_timing both before and after set_propagation_clock the one before set_propagation_clock is unnecessary. One update_timing command is in pt.tcl – the other is in a constraint file orca_pt_other.tcl) sourced by orca_pt_constraints which is sourced by pt.tcl, and it reduce the CPU timing consumption.

Q: Why might the exit command output be a good place to start before reviewing your log file?

A: It gives a high-level summary of potential trouble spots, message both warnings and information, timing updates and performances statistics.

5. Analyze STA Reports:

- Study timing reports generated by PrimeTime.
- Learn to identify slack values, critical paths, and timing violations.
- Forms the basis for debugging and timing closure.
- Execute the following to display the clock in ORCA .

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01

File Edit View Search Terminal Help

pt_shell> report_clock

Report : clock
Design : ORCA
Version: T-2022.03-SP4
Date : Mon Oct 6 18:09:20 2025

Attributes:

p - Propagated clock
G - Generated clock
I - Inactive clock

Clock	Period	Waveform	Attrs	Sources
PCI_CLK	15.000	{0.000 7.500}	p	{pclk}
SDRAM_CLK	7.500	{0.000 3.750}	p	{sdr_clk}
SD_DDR_CLK	7.500	{0.000 3.750}	p, G	{sd_CK}
SD_DDR_CLKn	7.500	{3.750 7.500}	p, G	{sd_CKn}
SYS_2x_CLK	4.000	{0.000 2.000}	p, G	{I_CLOCK_GEN/I_CLKMUL/CLK_2X}
SYS_CLK	8.000	{0.000 4.000}	p	{sys_clk}

Generated Clock	Master Source	Generated Source	Master Clock	Waveform Modification
-----------------	---------------	------------------	--------------	-----------------------

```

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help
p - Propagated clock
G - Generated clock
I - Inactive clock

Clock      Period      Waveform      Attrs      Sources
-----
PCI_CLK    15.000      {0.000 7.500} p          {pclk}
SDRAM_CLK  7.500       {0.000 3.750} p          {sdr_clk}
SD_DDR_CLK 7.500       {0.000 3.750} p, G      {sd_clk}
SD_DDR_CLKn 7.500      {3.750 7.500} p, G      {sd_CKn}
SYS_2x_CLK 4.000       {0.000 2.000} p, G      {I_CLOCK_GEN/I_CLKMUL/CLK_2X}
SYS_CLK    8.000       {0.000 4.000} p          {sys_clk}

Generated Clock      Master Source      Generated Source      Master Clock      Waveform Modification
-----
SD_DDR_CLK          I_CLOCK_GEN/I_PLL_SD/CLK      sd_CK          SDRAM_CLK          mult(1)
SD_DDR_CLKn         I_CLOCK_GEN/I_PLL_SD/CLK      sd_CKn         SDRAM_CLK          mult(1), inv
SYS_2x_CLK           sys_clk          I_CLOCK_GEN/I_CLKMUL/CLK_2X      SYS_CLK          mult(2)

1
pt_shell>

```



- Create a single, “short” timing report for setup for the clock SYS_CLK. Use command-line expansion(the tab key) to expand both the command AND the options-group and -path.

```
pt_shell> report_timing -group SYS_CLK -path short
```

```
*****
```

```
Report : timing
```

```
-path_type short  
-delay_type max  
-max_paths 1  
-group SYS_CLK  
-sort_by slack
```

```
Design : ORCA
```

```
Version: T-2022.03-SP4
```

```
Date : Mon Oct 6 18:13:48 2025
```

```
*****
```

```
Startpoint: I_ORCA_TOP/I_BLENDER/s3_op2_reg[18]  
(rising edge-triggered flip-flop clocked by SYS_CLK)
```

```
Endpoint: I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]  
(rising edge-triggered flip-flop clocked by SYS_CLK)
```

```
Last common pin: I_ORCA_TOP/I_BLENDER/bufbdfG682I8/Z
```

```
Path Group: SYS_CLK
```

```
Path Type: max
```

```
Min Clock Paths Derating Factor : 0.900
```

Point	Incr	Path
clock SYS_CLK (rise edge)	0.000	0.000
clock network delay (propagated)	3.247	3.247

Oct 6 18:14

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01

File Edit View Search Terminal Help

Min Clock Paths Derating Factor : 0.900

Point	Incr	Path
clock SYS_CLK (rise edge)	0.000	0.000
clock network delay (propagated)	3.247	3.247
I_ORCA_TOP/I_BLENDER/s3_op2_reg[18]/CP (sdnrb1)	0.000	3.247 r
I_ORCA_TOP/I_BLENDER/s3_op2_reg[18]/Q (sdnrb1)	0.516 &	3.763 r
...		
I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]/D (sdnrb1)	8.242 &	12.004 f
data arrival time		12.004
clock SYS_CLK (rise edge)	8.000	8.000
clock network delay (propagated)	2.736	10.736
clock reconvergence pessimism	0.470	11.206
I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]/CP (sdnrb1)		11.206 r
library setup time	-0.189	11.017
data required time		11.017
data required time		11.017
data arrival time		-12.004
slack (VIOLATED)		-0.987

1
pt_shell>

Red Hat
Enterprise Linux

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01

File Edit View Search Terminal Help

```
pt_shell> !rep -delay min
report_timing -group SYS_CLK -path short -delay min
*****
```

Report : timing

```
-path_type short
-delay_type min
-max_paths 1
-group SYS_CLK
-sort_by slack
```

Design : ORCA

Version: T-2022.03-SP4

Date : Mon Oct 6 18:16:09 2025

```
Startpoint: I_ORCA_TOP/I_PARSER/out_bus_reg[10]
(rising edge-triggered flip-flop clocked by SYS_CLK)
```

```
Endpoint: I_ORCA_TOP/I_BLENDER/rem_green_reg
(rising edge-triggered flip-flop clocked by SYS_CLK)
```

```
Last common pin: I_CLK_SOURCE_SYS_CLK/Z
```

Path Group: SYS_CLK

Path Type: min

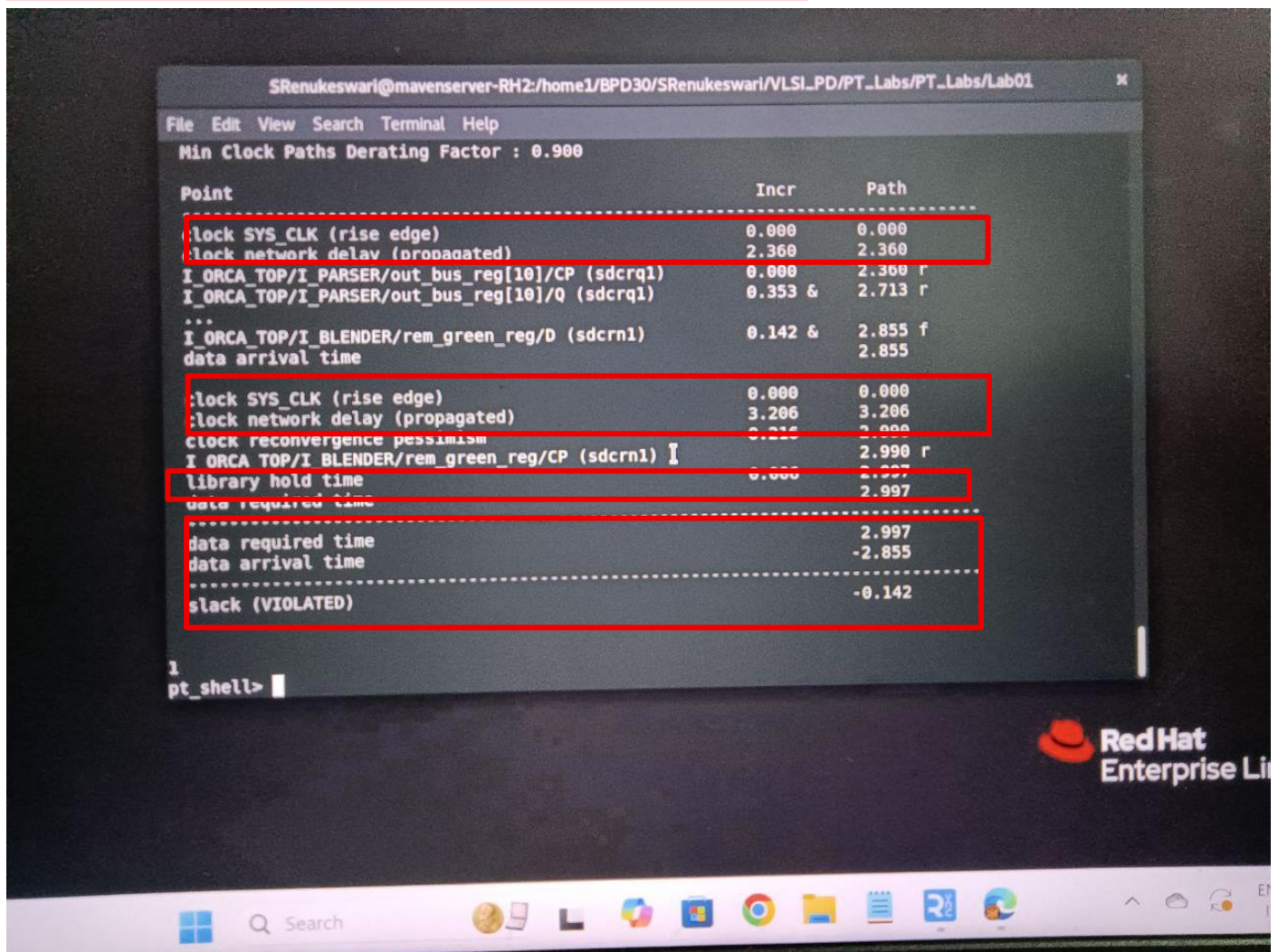
Min Data Paths Derating Factor : 0.900

Min Clock Paths Derating Factor : 0.900

Incr

Path

Point



- 4 Clues that we identify the report is for setup or hold:
- Path type max for setup and min for hold
- CLK rise edge is two successive points it is called setup and both clk are rise at 0.00 then it is hold.
- Library is setup time for setup and hold time for hold.
- Slack = data required time – data arrival time for both.

```

SRenukeswari@mavenserver-RH2:/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/Lab01
File Edit View Search Terminal Help

Point                                Incr    Path
-----
clock SYS_CLK (rise edge)            0.000    0.000
clock network delay (propagated)     3.247    3.247
I_ORCA_TOP/I_BLENDER/s3_op2_reg[18]/CP (sdnrb1) 0.000    3.247 r
I_ORCA_TOP/I_BLENDER/s3_op2_reg[18]/Q (sdnrb1) 0.516 &  3.763 r
...
I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]/D (sdnrb1) 8.242 &  12.004 f
data arrival time                    12.004

clock SYS_CLK (rise edge)            8.000    8.000
clock network delay (propagated)     2.736    10.736
clock reconvergence pessimism        0.470    11.206
I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]/CP (sdnrb1) 11.206 r
library setup time                   -0.189    11.017
data required time                   11.017

data required time                    11.017
data arrival time                    -12.004

slack (VIOLATED)                     -0.987

1
pt_shell> !rep -delay min
report_timing -group SYS_CLK -path short -delay min

```

Here some questions related to this Task:

Q: How Many clock are in ORCA?

A: 6 Clock, there are:

PCI_CLK, SDRAM_CLK, SD_DDR_CLK, SD_DDR_CLKn, SYS_2x_CLK, SYS_CLK.

Q: There are at least 4 clues that this report is for setup and not for hold. How many Can you identify?

A: 4 Clues that we identify the report is for setup :

- Path type max for setup.
- CLK rise edge is two successive points it is called setup.
- Library is setup time for setup.
- Slack = data required time – data arrival time for setup.

Q: Identify the instance names of the start and end point flip-flops?

A: Yes, at header section in timing report, we identify the startpoint and endpoint.

Q: The clock skew for this timing path is 0.511ns; which two lines in the report can you use to calculate this?

A: we use setup report for this, in that we take clock network delay (propagated) for two rising edges values is 3.247 and second one is 2.736.

$$\begin{aligned}\text{Clock Skew} &= \text{clock network delay (propagation) 1} - \text{clock network delay (propagation)2} \\ &= 3.247 - 2.736\end{aligned}$$

$$\text{Clock Skew} = 0.511\text{ns}$$

Q: How does this clock skew affect slack (i.e., does the clock skew help or hurt slack)?

A: The clock skew hurts the slack for setup in this specific timing report. The clock latency to the start point flip-flop causes the data to arrive 0.511ns later causing a large setup timing violations.

Q: How large is this violation in comparison to the clock period?

A: The period is 8ns. The violation is 0.987. It is approximately 12% of the clock period.

$$\begin{aligned}\text{Clock period} &= (\text{Violation/clock rise edge}) \times 100 \\ &= (0.987/8) \times 100\end{aligned}$$

$$\text{Clock period} = 12\%$$

Q: There are at least 4 clues that indicate this is a hold report and not a setup report. How many can you find?

A: 4 Clues that we identify the report is for setup or hold:

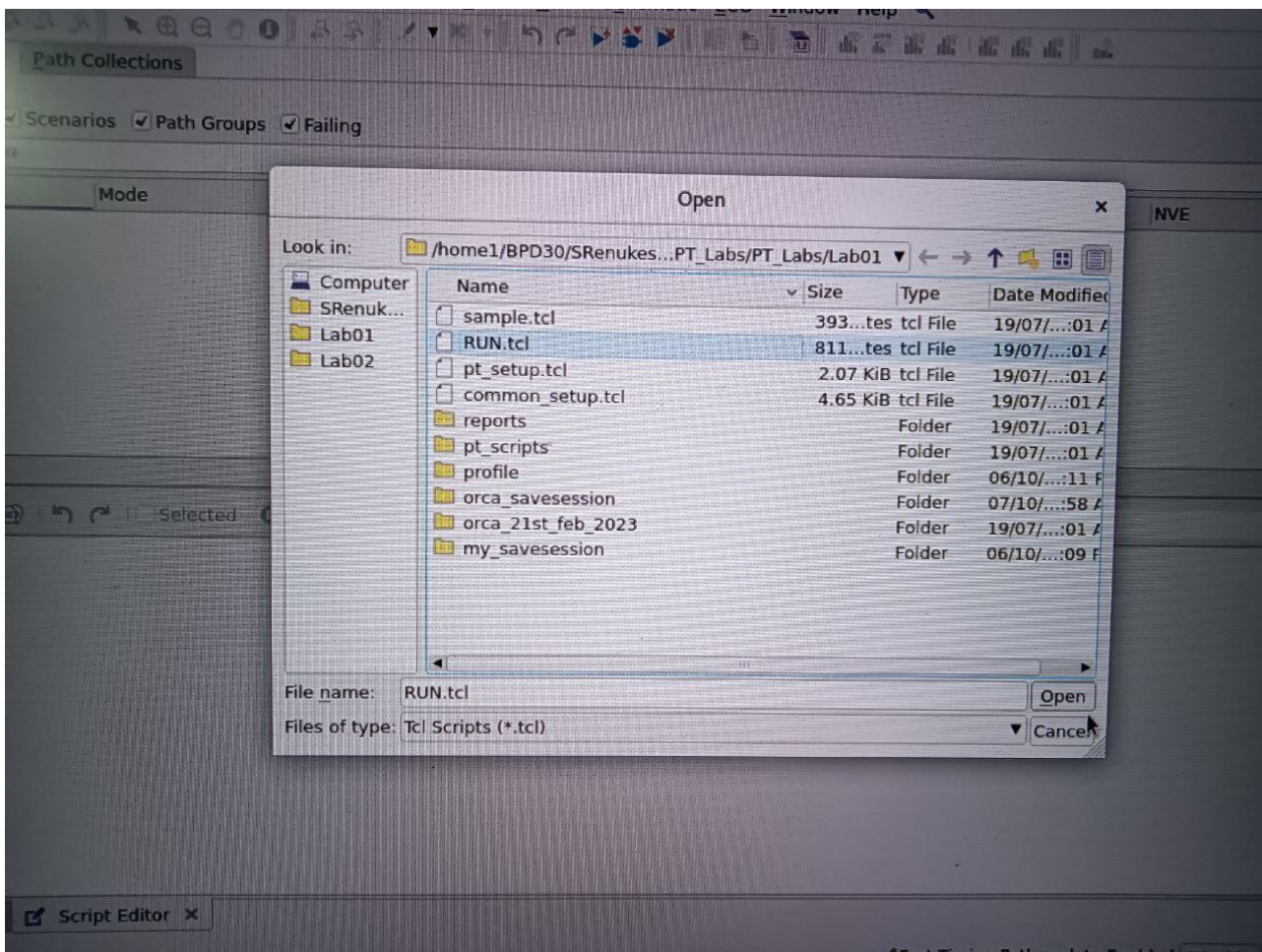
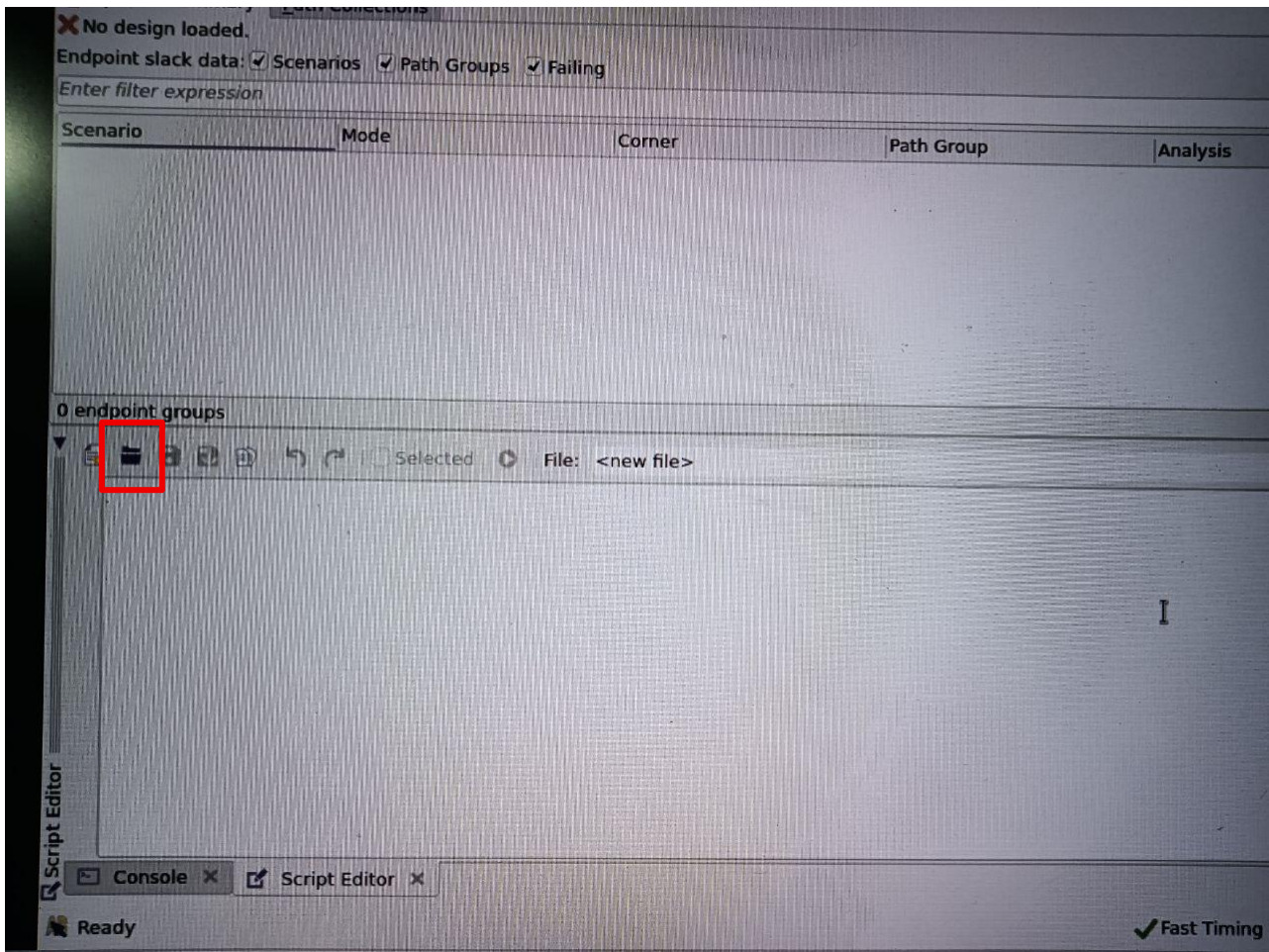
- Path type min for hold
- CLK rise edge of both clocks are at 0.00 then it is hold.
- Library is hold time for hold.
- Slack = data required time – data arrival time for hold.

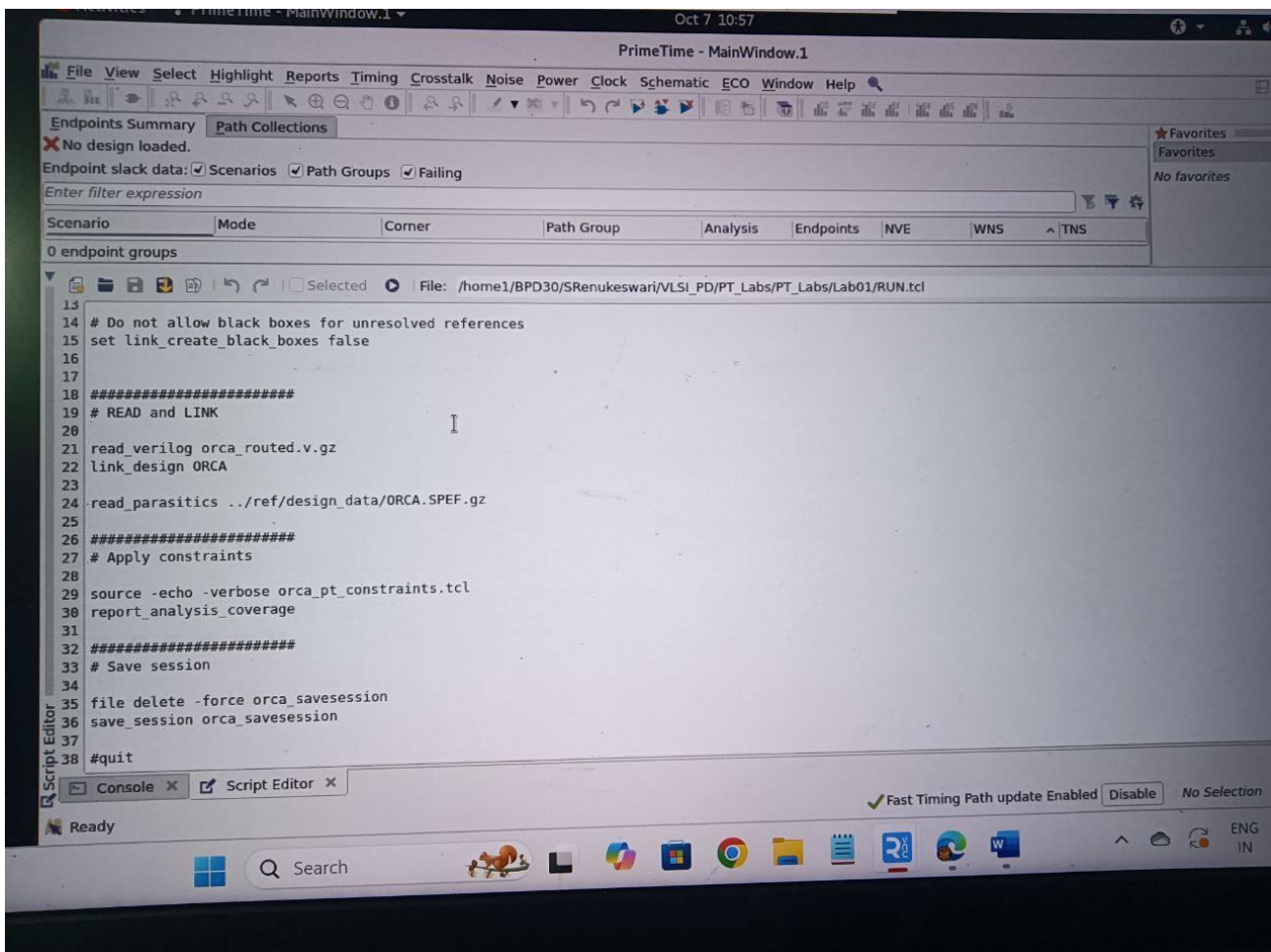
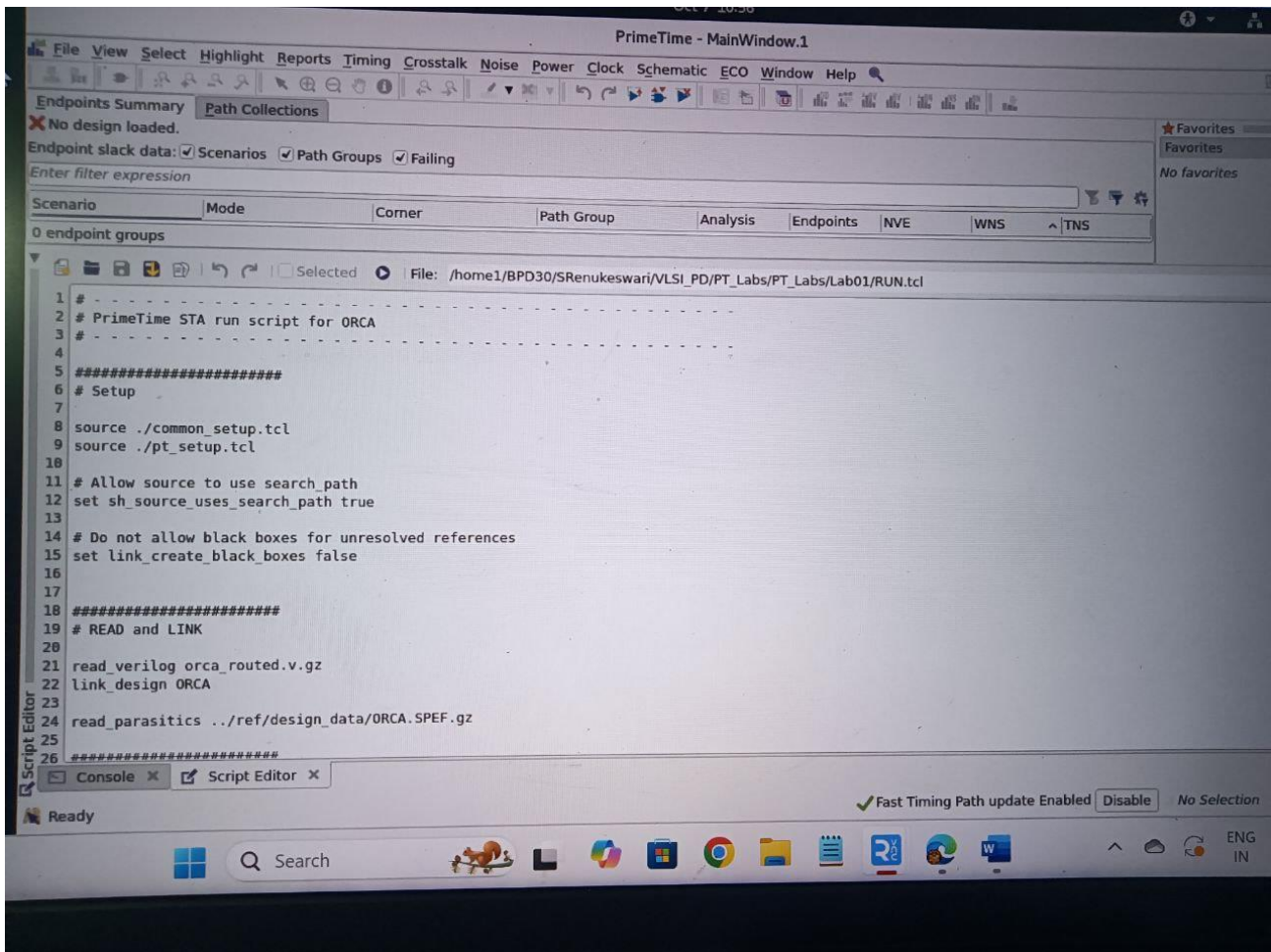
Q: How does the clock skew in this hold report affect slack (i.e., does the clock skew help or hurt slack)?

A: From the report, you can see (after applying the clock network delay) that the capture clock edge arrives later than the launch clock edge. This hurts the hold slack causing a violation.

Here we can see the GUI interface and graph for slack:

- Enter into a Prime time tool Interface, and open RUN.tcl file in this file setup , Read and Link, Apply constraints, and save session is there. Each session now we RUN by selecting the cmd lines and click on RUN Symbol and output we see in console window.





- Now see output in console window for each session.

The screenshot shows the Synopsys IDE interface. At the top, there are tabs for 'Endpoints Summary' and 'Path Collections'. Below these, a status bar indicates 'Timing is up to date' and 'Endpoint slack data: Scenarios Path Groups Failing'. A table of values is visible, showing various numerical data points. The main console window displays the following text:

```

Error: Library Compiler executable path is not set. (PT-063)

Information: Completed sourcing .synopsys_pt.setup

pt_shell> gui_start
pt_shell>
# Setup
source ./common_setup.tcl
VDD
VDD
source ./pt_setup.tcl
# Allow source to use search_path
set sh_source_uses_search_path true
true
# Do not allow black boxes for unresolved references
set link_create_black_boxes falsefalse
# READ and LINK
read_verilog orca_routed.v.gz
1
link_design ORCA
Loading verilog file '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/design_data/orca_routed.v.gz'

```

The console window has tabs for 'Log' and 'History'. Below the console, there are buttons for 'Console' and 'Script Editor'. The status bar at the bottom shows 'Ready' and 'Fast Timing P'.

Endpoints Summary Path Collections

✓ Timing is up to date Update
Endpoint slack data: ✓ Scenarios ✓ Path Groups ✓ Failing

16 endpoints

1
link_design ORCA
Loading verilog file '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/design_data/orca_routed.v.gz'
Loading db file '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/sc_max.db'
Loading db file '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/io_max.db'
Loading db file '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/special.db'
Loading db file '/home1/BPD30/SRenukeswari/VLSI_PD/PT_Labs/PT_Labs/ref/libs/mem_max.db'
Linking design ORCA...
Information: Removing 24 unneeded designs..... (LNK-034)
Information: 439 (81.75%) library cells are unused in library cb13fs120_tsmc_max..... (LNK-045)
Information: 82 (95.35%) library cells are unused in library cb13io320_tsmc_max..... (LNK-045)
Information: total 521 library cells are unused (LNK-046)
Design 'ORCA' was successfully linked.
Information: There are 20403 leaf cells, ports, hiers and 24967 nets in the design (LNK-047)

1
read_parasitics ../ref/design_data/ORCA.SPEF.gz
Information: Log for 'read_parasitics command' will be generated in 'parasitics_command.log'. (PARA-107)

1
Apply constraints
source -echo -verbose orca_pt_constraints.tcl
#

Log History

pt_shell>

Console x Script Editor x

Ready

✓ Fast Timing Path up

Endpoints Summary

Path Collections

✓ Timing is up to date

Endpoint slack data: ✓

Scenarios

✓ Path Groups

✓ Failing

-3.67 -3.665 -3.66 -3.655 -3.65 -3.645 -3.64 -3.635 -3.63 -3.625 -3.62 -3.615 -3.61 -3.605 -3.6 -3.595 -3.59 -3.585 -3.58 -3.57

L1/E1 IN/OUT EXPRESSIVE

16 endpoints

Report Worst Path | Insert Worst Paths | Create Paths

```
1
# Apply constraints
source -echo -verbose orca_pt_constraints.tcl
#
# Master constraints for the ORCA chip for functional mode
#
# Time Unit: ns
# Cap Unit : pf
source -echo -verbose orca_pt_clocks.tcl
#####
# Create functional clocks
#####
# PCI clock at 66 MHz
create_clock -period 15 -name PCI_CLK [get_ports pclk]
1
# System clock at 125 MHz
create_clock -period 8 -name SYS_CLK [get_ports sys_clk]
1
# sdram clock at 133 MHz
create_clock -period 7.5 -name SDRAM_CLK [get_ports sdr_clk]
1
```

Log

History

pt_shell>

Console x

Script Editor x

Ready

✓ Fast Timing Path u

Endpoints Summary

Path Collections

✓ Timing is up to date Update

Endpoint slack data: ✓ Scenarios ✓ Path Groups ✓ Failing

-3.67 -3.665 -3.66 -3.655 -3.65 -3.645 -3.64 -3.635 -3.63 -3.625 -3.62 -3.615 -3.61 -3.605 -3.6 -3.595 -3.59 -3.585 -3.58 -3.575

Literal Value Expression

16 endpoints

Report Worst Path Insert Worst Paths Create Path

```
create_clock -period 8 -name SYS_CLK [get_ports sys_clk]
1
# sdram clock at 133 MHz
create_clock -period 7.5 -name SDRAM_CLK [get_ports sdr_clk]
1
#create_clock -p 7.5 -name SD_DDR_CLK [get_pin I_CLOCK_GEN/I_PLL_SD/CLK]
# Create outgoing clocks for DDR interface
#create_generated_clock \
    -source [get_ports sdr_clk] \
    -multiply_by 1 \
    -name SD_DDR_CLK \
    [get_ports sd_CK]
#15 aug 07 -- move source from port to pll pin
create_generated_clock \
    -source [get_pins I_CLOCK_GEN/I_PLL_SD/CLK] \
    -multiply_by 1 \
    -name SD_DDR_CLK \
    [get_ports sd_CK]
1
#15 aug 07 -- move source from port to pll pin
#create_generated_clock \
```

Log

History

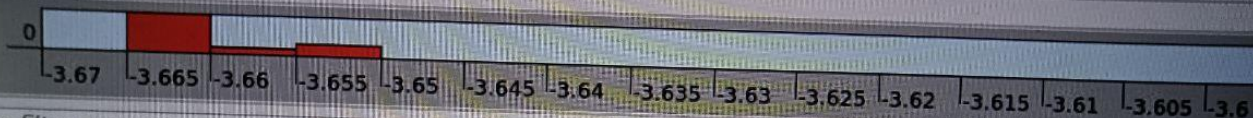
pt_shell>

Console x

Script Editor x

✓ Fast Timing

TX endpoint groups



Enter filter expression

slack ^ object class full name

16 endpoints

Report Worst Path

```
[get_ports sd_CK]
1
#15 aug 07 -- move source from port to pll pin
#create_generated_clock \
  -source [get_ports sdr_clk] \
  -multiply_by 1 -invert \
  -name SD_DDR_CLKn \
  [get_ports sd_CKn]
create_generated_clock \
  -source [get_pins I_CLOCK_GEN/I_PLL_SD/CLK] \
  -multiply_by 1 -invert \
  -name SD_DDR_CLKn \
  [get_ports sd_CKn]
1
# Create internally derived clocks
create_generated_clock \
  -source [get_ports sys_clk] \
```

Log

History

pt_shell>

Console x

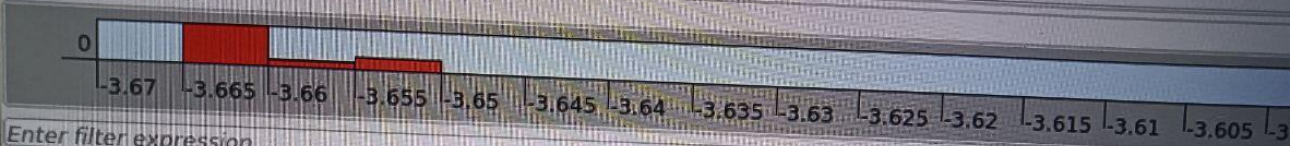
Script Editor x

Ready

Fast Time

1x Endpoint groups

0



Enter filter expression

slack ^ object class full name

16 endpoints

Report Worst Pa

```
# Create internally derived clocks
create_generated_clock \
  -source [get_ports sys_clk] \
  -multiply_by 2 \
  -name SYS_2x_CLK \
  [get_pins I_CLOCK_GEN/I_CLKMUL/CLK_2X]

1
# Setup asynchronous clock groups
set_clock_groups -name func_async -asynchronous \
  -group [get_clocks PCI_CLK] \
  -group [get_clocks SD*] \
  -group [get_clocks SYS*]

1
1
1
1
source -echo -verbose orca_pt_ioconst.tcl
# Constrain interface
```

Log

History

pt_shell>

Console x

Script Editor x

Ready

✓ Fast

Activities PrimeTime - MainWindow.1 Oct 7 11:07

PrimeTime - MainWindow.1 - ORCA

File View Select Highlight Reports Timing Crosstalk Noise Power Clock Schematic ECO Window Help

Endpoints Summary Path Collections

Timing is up to date. Update

Endpoint slack data: Scenarios Path Groups Failing

16 endpoints remaining

slack 16 endpoints

Report Worst Path Inspect Worst Paths Create Paths ... Timing

```
source -echo -verbose orca_pt_ioconst.tcl
# Constrain interface
set pci_io_ports [get_ports p* -filter "port_direction==inout"]
{"pdevsel_n", "pstop_n", "pperr_n", "pserr_n", "ppar", "pc_be[3]", "pc_be[2]", "pc_be[1]", "pc_be[0]", "pframe_n", "ptrdy_n", "pirdy_n", "pad[15]", "pad[14]", "pad[13]", "pad[12]", "pad[11]", "pad[10]", "pad[9]", "pad[8]", "pad[7]", "pad[6]", "pad[5]", "pad[4]", "pad[3]", "pad[2]", "pad[1]", "pad[0]}
set pci_in_ports [add_to_collection [get_ports "pidsel pgnt_n pm66en"] $pci_io_ports]
{"pidsel", "pgnt_n", "pm66en", "pdevsel_n", "pstop_n", "pperr_n", "pserr_n", "ppar", "pc_be[3]", "pc_be[2]", "pc_be[1]", "pc_be[0]", "pframe_n", "ptrdy_n", "pirdy_n", "pad[15]", "pad[14]", "pad[13]", "pad[12]", "pad[11]", "pad[10]", "pad[9]", "pad[8]", "pad[7]", "pad[6]", "pad[5]", "pad[4]", "pad[3]", "pad[2]", "pad[1]", "pad[0]}
set pci_out_ports [add_to_collection [get_ports "pseq_n"] $pci_io_ports]
{"pseq_n", "pdevsel_n", "pstop_n", "pperr_n", "pserr_n", "ppar", "pc_be[3]", "pc_be[2]", "pc_be[1]", "pc_be[0]", "pframe_n", "ptrdy_n", "pirdy_n", "pad[15]", "pad[14]", "pad[13]", "pad[12]", "pad[11]", "pad[10]", "pad[9]", "pad[8]", "pad[7]", "pad[6]", "pad[5]", "pad[4]", "pad[3]", "pad[2]", "pad[1]", "pad[0]}
set pci_ports [add_to_collection $pci_in_ports $pci_out_ports]
{"pidsel", "pgnt_n", "pm66en", "pdevsel_n", "pstop_n", "pperr_n", "pserr_n", "ppar", "pc_be[3]", "pc_be[2]", "pc_be[1]", "pc_be[0]", "pframe_n", "ptrdy_n", "pirdy_n", "pad[15]", "pad[14]", "pad[13]", "pad[12]", "pad[11]", "pad[10]", "pad[9]", "pad[8]", "pad[7]", "pad[6]", "pad[5]", "pad[4]", "pad[3]", "pad[2]", "pad[1]", "pad[0]}
set sd_io_ddr_ports [get_ports "sd_DQ*"]
{"sd_DQ[15]", "sd_DQ[14]", "sd_DQ[13]", "sd_DQ[12]", "sd_DQ[11]", "sd_DQ[10]", "sd_DQ[9]", "sd_DQ[8]", "sd_DQ[7]", "sd_DQ[6]", "sd_DQ[5]", "sd_DQ[4]", "sd_DQ[3]", "sd_DQ[2]", "sd_DQ[1]", "sd_DQ[0]}
set sd_out_ports [remove_from_collection [get_ports sd_*] \
  {"sd_DQ[15]", "sd_DQ[14]", "sd_DQ[13]", "sd_DQ[12]", "sd_DQ[11]", "sd_DQ[10]", "sd_DQ[9]", "sd_DQ[8]", "sd_DQ[7]", "sd_DQ[6]", "sd_DQ[5]", "sd_DQ[4]", "sd_DQ[3]", "sd_DQ[2]", "sd_DQ[1]", "sd_DQ[0]}]
  {"sd_BWS[1]", "sd_BWS[0]", "sd_LD", "sd_RW", "sd_A[9]", "sd_A[8]", "sd_A[7]", "sd_A[6]", "sd_A[5]", "sd_A[4]", "sd_A[3]", "sd_A[2]", "sd_A[1]", "sd_A[0]}
set sd_ports [add_to_collection $sd_io_ddr_ports $sd_out_ports]
{"sd_DQ[15]", "sd_DQ[14]", "sd_DQ[13]", "sd_DQ[12]", "sd_DQ[11]", "sd_DQ[10]", "sd_DQ[9]", "sd_DQ[8]", "sd_DQ[7]", "sd_DQ[6]", "sd_DQ[5]", "sd_DQ[4]", "sd_DQ[3]", "sd_DQ[2]", "sd_DQ[1]", "sd_DQ[0]", "sd_BWS[1]", "sd_BWS[0]", "sd_LD", "sd_RW", "sd_A[9]", "sd_A[8]", "sd_A[7]", "sd_A[6]", "sd_A[5]", "sd_A[4]", "sd_A[3]", "sd_A[2]", "sd_A[1]", "sd_A[0]}
```

Log History

pt_shell>

Console Script Editor

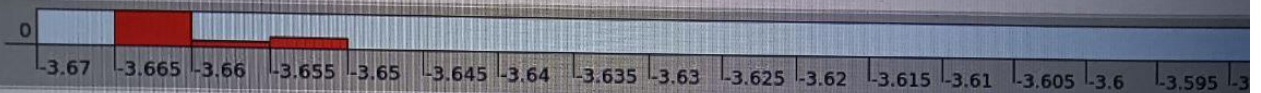
Ready Fast Timing Path update Enabled Disable No Selection

07-

✓ Timing is up to date. Update

Endpoint slack data: ☒ Scenarios ☒ Path Groups ☒ Failing

1x endpoint groups



Enter filter expression

slack ^ object class full name

16 endpoints

Report Worst Path Inspect

```
{"sd_BWS[1]", "sd_BWS[0]", "sd_LD", "sd_RW", "sd_A[9]", "sd_A[8]", "sd_A[7]", "sd_A[6]", "sd_A[5]", "sd_A[4]", "sd_A[3]", "sd_A[2]", "sd_A[1]", "sd_A[0]", "sd_DQ[15]", "sd_DQ[14]", "sd_DQ[13]", "sd_DQ[12]", "sd_DQ[11]", "sd_DQ[10]", "sd_DQ[9]", "sd_DQ[8]", "sd_DQ[7]", "sd_DQ[6]", "sd_DQ[5]", "sd_DQ[4]", "sd_DQ[3]", "sd_DQ[2]", "sd_DQ[1]", "sd_DQ[0]"}
set sd_ports [add_to_collection $sd_io_ddr_ports $sd_out_ports]
{"sd_DQ[15]", "sd_DQ[14]", "sd_DQ[13]", "sd_DQ[12]", "sd_DQ[11]", "sd_DQ[10]", "sd_DQ[9]", "sd_DQ[8]", "sd_DQ[7]", "sd_DQ[6]", "sd_DQ[5]", "sd_DQ[4]", "sd_DQ[3]", "sd_DQ[2]", "sd_DQ[1]", "sd_DQ[0]"}
# Constrain PCI ports
set_input_delay 8.000 -max -clock PCI_CLK -add_delay $pci_in_ports
1
set_input_delay 2.000 -min -clock PCI_CLK -add_delay $pci_in_ports
1
set_output_delay 4.000 -max -clock PCI_CLK -add_delay $pci_out_ports
1
set_output_delay -1.000 -min -clock PCI_CLK -add_delay $pci_out_ports
1
# Constrain SDRAM ports
set_input_delay 0.800 -max -add_delay -clock SDRAM_CLK $sd_io_ddr_ports
1
set_input_delay 0.800 -max -add_delay -clock SDRAM_CLK -clock_fall $sd_io_ddr_ports
1
```

Console

Log

History

pt_shell>

Console x

Script Editor x

Ready

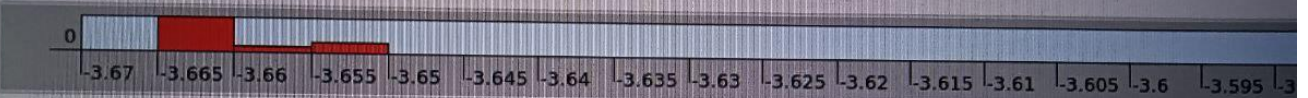
✓ Fast Timing F

Endpoints Summary Path Collections

✓ Timing is up to date. Update

Endpoint slack data: ✓ Scenarios ✓ Path Groups ✓ Failing

1x endpoint groups



Enter filter expression

slack ^ object class full name

16 endpoints

Report Worst Path Inspect

```
# Constrain SDRAM ports
set_input_delay 0.800 -max -add_delay -clock SDRAM_CLK $sd_io_ddr_ports
1
set_input_delay 0.800 -max -add_delay -clock SDRAM_CLK -clock_fall $sd_io_ddr_ports
1
set_input_delay 0.200 -min -add_delay -clock SDRAM_CLK $sd_io_ddr_ports
1
set_input_delay 0.200 -min -add_delay -clock SDRAM_CLK -clock_fall $sd_io_ddr_ports
1
# Constrain DDR interface
set_output_delay 0.750 -max -add_delay -clock SD_DDR_CLK $sd_ports
1
set_output_delay 0.750 -max -add_delay -clock SD_DDR_CLK -clock_fall $sd_io_ddr_ports
1
set_output_delay -0.100 -min -add_delay -clock SD_DDR_CLK $sd_ports
1
set_output_delay -0.100 -min -add_delay -clock SD_DDR_CLK -clock_fall $sd_io_ddr_ports
```

Console

Log

History

pt_shell>

Console x

Script Editor x

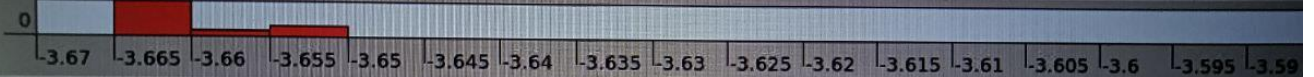
Ready

✓ Fast Timing

✓ Timing is up to date. Update ▾

Endpoint slack data: ☒ Scenarios ☒ Path Groups ☒ Failing

1x endpoint arrives



Enter filter expression

slack ^ object class full name

16 endpoints

Report Worst Path Inspect Worst Path

```
1
set_output_delay -0.100 -min -add_delay -clock SD_DDR_CLK -clock_fall $sd_io_ddr_ports
1
# Set up constraints for clock gating in Blender design
set_clock_gating_check -setup 0.200 -hold 0.200 [get_clocks SYS_CLK]
1
1
1
source -echo -verbose orca_pt_other.tcl
#####
# Other special constraints for ORCA
#####
#####
# Setup case analysis values
set_case_analysis 0 [get_ports "test_mode scan_en power_save"]
1
set_case_analysis 1 [get_ports "pm66en prst_n"]
```

Log

History

pt_shell>

Console x

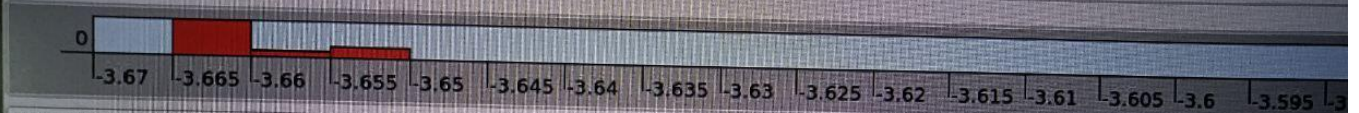
Script Editor x

Ready

✓ Fast Timing Path

Endpoint slack data: ☒ Scenarios ☒ Path Groups ☒ Failing

1x endpoint groups



Enter filter expression

slack ^ object class full name

16 endpoints

Report Worst Path Inspect

```
1
set_case_analysis 1 [get_ports "pm66en prst_n"]
1
#####
# Small derating factor
set_timing_derate -early 0.9
1
#for RC-005, port is a 1.08 volt, but pad side of pad cell is at 3 volt
create_operating_conditions -name 3V -library cb13io320_tsmc_max -voltage 3 -process 1 -temperature 125
1
set_operating_conditions 3V -object_list [get_ports *]
1
#for check_timing complaint about undriven nets having parasitics
set_driving_cell -lib_cell inv0d1 [all_inputs]
1
#to lower pad cell delays
set_input_transition 0.1 sd_DQ
```

Log

History

pt_shell>

Console

Console x

Script Editor x

Ready

Fast Timing Pa



Search



File View Select Highlight Reports Timing Crosstalk Noise Power Clock Schematic ECO Window Help

Endpoints Summary Path Collections

Timing is up to date. Update

Endpoint slack data: ☒ Scenarios ☒ Path Groups ☒ Failing

16 endpoints

Enter filter expression

slack ^ object class full name

Report Worst Path Inspect Worst Paths Create Paths ... Timing

```
1
set_operating_conditions 3V -object_list [get_ports *]
1
# for check timing complaint about undriven nets having parasitics
set_driving_cell -lib_cell inv0d1 [all_inputs]
1
# to lower pad cell delays
set_input_transition 0.1 sd_DQ
1
# Setup gating checks for DDR interface
set_all_I0_mux_pins [get_pins I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out */I0]
{"I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_10/I0", "I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_11/I0", "I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_12/I0", "I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_13/I0", "I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_14/I0", "I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_15/I0"}
set_all_I1_mux_pins [get_pins I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out */I1]
{"I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_10/I1", "I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_11/I1", "I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_12/I1", "I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_13/I1", "I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_14/I1", "I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_15/I1"}
set_clock_gating_check -low -setup 0 -hold 0 $all_I0_mux_pins
1
set_clock_gating_check -high -setup 0 -hold 0 $all_I1_mux_pins
```

Log History

pt_shell>

Console x Script Editor x

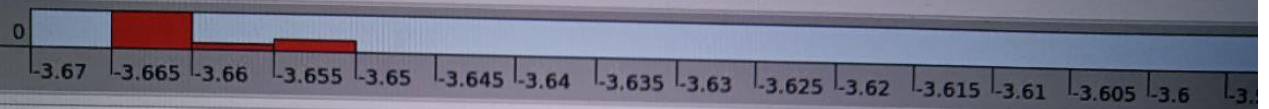
Ready

Fast Timing Path update Enabled Disable No Selection

Search

ENG IN

1x Endpoint timing



Enter filter expression

slack ^ | object class | full name

16 endpoints

Report Worst Path

```
set_clock_gating_check -low -setup 0 -hold 0 $all_I0_mux_pins
1
set_clock_gating_check -high -setup 0 -hold 0 $all_I1_mux_pins
1
# Stop the clock gating data signal from propagating
set_timing_clock_gating_propagate_enable false
false
#####
# Set a false path for all sd_DQ[0-31] output pad enable paths
for {set i 0} {$i < 16} {incr i} {
    set_false_path -from [get_pins I_ORCA_TOP/I_SDRAM_IF/sd_DQ_en_reg[$i]/CP] \
        -through [get_pins sdram_DQ_iopad_$i/OEN]
}
#####
# Account for negative latency (phase shift) due to PLL
# SDRAM_CLK and PCI_CLK
PLL_SHIFT I_CLOCK_GEN/I_PLL_PCI_FB_CLK CLK
```

Log

History

pt_shell>

Console

Script Editor

Ready

Fast Time

Oct 7 11:11

PrimeTime - MainWindow.1 - ORCA

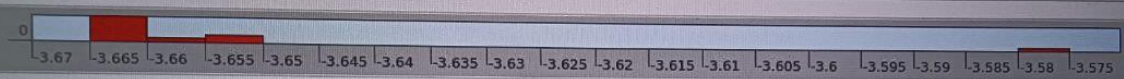
File View Select Highlight Reports Timing Crosstalk Noise Power Clock Schematic ECO Window Help

Endpoints Summary Path Collections

Timing is up to date. Update

Endpoint slack data: ☒ Scenarios ☒ Path Groups ☒ Failing

IN endpoint timing



Attribute slack

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Enter filter expression

slack ^ | object class | full name

16 endpoints

Report Worst Path

Inspect Worst Paths

Create Paths ...

Timing

SDRAM_CLK and PCI_CLK

PLL_SHIFT I_CLOCK_GEN/I_PLL_PCI FB_CLK CLK

1

PLL_SHIFT I_CLOCK_GEN/I_PLL_SD FB_CLK CLK

1

update_timing -full

Information: Building multi voltage information for entire design. (MV-022)

Warning: Some timing arcs have been disabled for breaking timing loops or because of constant propagation. Use the 'report_disable_timing' command to get

Information: Inferring 33 clock-gating checks. (PTE-017)

Information: Expanding clock 'SYS_2x_CLK' to base period of 8.000 (old period was 4.000, added 2 edges). (PTE-016)

1

1

1

1

1

1

report_analysis_coverage

Log History

pt_shell>

Console x

Script Editor x

Ready

Fast Timing Path update Enabled

Disable

No Selection

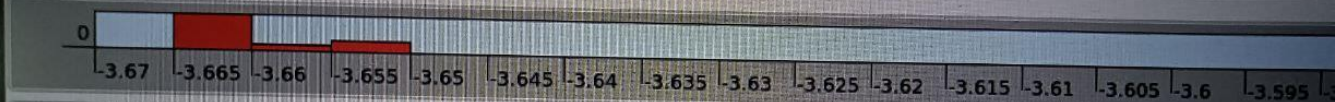


Search



Endpoint slack data: ☒ Scenarios ☒ Path Groups ☒ Failing

1X endpoint groups



Enter filter expression

slack ^ object class full name

16 endpoints

Report Worst Path

▼ report_analysis_coverage

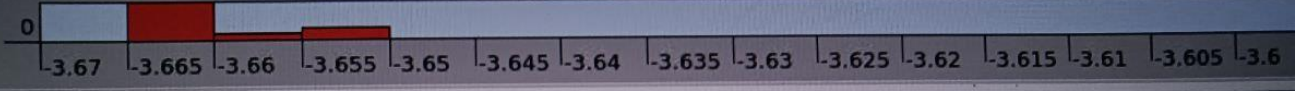
Report : analysis_coverage
Design : ORCA
Version: T-2022.03-SP4
Date : Tue Oct 7 11:05:18 2025

Type of Check	Total	Met	Violated	Untested
setup	9629	3565 (37%)	23 (0%)	6041 (63%)
hold	9629	3535 (37%)	53 (1%)	6041 (63%)
recovery	1316	1210 (92%)	0 (0%)	106 (8%)
removal	1316	1210 (92%)	0 (0%)	106 (8%)
min_period	20	20 (100%)	0 (0%)	0 (0%)
min_pulse_width	7273	5957 (82%)	0 (0%)	1316 (18%)
clock_gating_setup	33	33 (100%)	0 (0%)	0 (0%)

Log History
pt_shell>
Console x Script Editor x

Ready ✓ Fast Timing P





Enter filter expression

slack ^ | object class | full name
16 endpoints

Report Worst Path

Type of Check	Total	Met	Violated	Untested
setup	9629	3565 (37%)	23 (0%)	6041 (63%)
hold	9629	3535 (37%)	53 (1%)	6041 (63%)
recovery	1316	1210 (92%)	0 (0%)	106 (8%)
removal	1316	1210 (92%)	0 (0%)	106 (8%)
min_period	20	20 (100%)	0 (0%)	0 (0%)
min_pulse_width	7273	5957 (82%)	0 (0%)	1316 (18%)
clock_gating_setup	33	33 (100%)	0 (0%)	0 (0%)
clock_gating_hold	33	33 (100%)	0 (0%)	0 (0%)
out_setup	75	43 (57%)	32 (43%)	0 (0%)
out_hold	75	75 (100%)	0 (0%)	0 (0%)
All Checks	29399	15681 (53%)	108 (0%)	13610 (46%)

Log

History

pt_shell>

Console x

Script Editor x

Ready

